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Department of Computer and Electronics Engineering

A FINAL REPORT ON

Nepali Sign Language Recognition for Emergency Phrases

Submitted in partial fulfillment of the requirements for the degree in Bachelor of Computer Engineering

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CERTIFICATE OF APPROVAL

This is to certify that this minor project work entitled **"Nepali Sign Language Recognition for Emergency Phrases"**, submitted by Asul Limbu (KCE080BCT005), Dipesh Singh (KCE080BCT008), Manee Das Shrestha (KCE080BCT013) and Sajan Sahikarmi (KCE080BCT035) has been examined and accepted as the partial fulfillment of the requirements for the degree of Bachelor in Computer Engineering. The project was carried out under special supervision and within the time frame prescribed by the syllabus.

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ABSTRACT

Nepali Sign Language (NSL) is the first language of Nepal's deaf community, yet very few hearing people understand it. The gap is most dangerous in an emergency, when a deaf person needs to convey an urgent message immediately and no interpreter is present. This project built a system that recognises six NSL emergency phrases from an ordinary camera and delivers each one as on-screen text and speech.

No public dataset of NSL emergency phrases exists, so one was recorded from scratch. The natural signed form of every phrase was established in person with Deaf experts at the Rehabilitation Centre for the Rural Deaf and the National Federation of the Deaf Nepal before any recording began. Eight signers then contributed 840 clips across seven classes: the six emergency phrases and a negative class covering non-signing motion.

Each frame is reduced by MediaPipe Holistic to 225 landmark coordinates instead of being stored as video, which keeps the representation compact and independent of background and lighting. Landmarks are normalised against two anatomical anchors, the shoulder midpoint for the body and the wrist for each hand, and every clip is standardised to 137 frames, the 95th percentile of the recorded frame-count distribution. A bidirectional Long Short-Term Memory network classifies the resulting sequence.

Evaluation used leave-one-signer-out folds, in which the model is scored only on a signer absent from its training data. Accuracy averaged 96.52% over eight folds and 96.67% pooled across all 840 predictions, with per-class F1 between 0.963 and 0.996 for the six emergency phrases. The same architecture reaches 99.21% under a random split, and that gap of 2.5 percentage points is a measure of how much a random split overstates performance on this kind of data. A seventh, negative class trained on non-signing motion lets the system stay silent instead of announcing an emergency phrase for arbitrary motion. The trained network was exported to TensorFlow Lite and deployed behind a streaming inference server, with an Android client that captures the sign, displays the recognised phrase, and speaks it aloud.

Keywords: *Nepali Sign Language, Emergency Phrases, Sign Language Recognition, MediaPipe Holistic, Hand Landmarks, BiLSTM, Signer-Independent Evaluation.*

Contents

List of Tables

List of Figures

List of Symbols and Abbreviation

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
APK	Android Package Kit
ASL	American Sign Language
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DFD	Data Flow Diagram
F1	F1 Score, the harmonic mean of precision and recall
FPS	Frames Per Second
GPU	Graphics Processing Unit
HMAC	Hash-Based Message Authentication Code
HTTP	Hypertext Transfer Protocol
ISL	Indian Sign Language
JPEG	Joint Photographic Experts Group
JSON	JavaScript Object Notation
LOSO	Leave-One-Signer-Out
LSTM	Long Short-Term Memory
MCP	Metacarpophalangeal (finger knuckle joint)
NFDN	National Federation of the Deaf Nepal
NSL	Nepali Sign Language
RCRD	Rehabilitation Centre for the Rural Deaf
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
SDLC	Software Development Life Cycle
TFLite	TensorFlow Lite
TTS	Text to Speech
UI	User Interface
WS	WebSocket
WSS	WebSocket Secure
YUV	Luminance and Chrominance Colour Encoding

CHAPTER 1
INTRODUCTION

1.1 Background Introduction

Sign languages are complete natural languages with their own vocabulary and grammar. They are not signed versions of spoken languages, and they do not translate word for word into them. Nepali Sign Language (NSL) is the language of the deaf community in Nepal and differs substantially from American, British, or Indian sign languages. It grew within Nepal's deaf community after the first deaf school opened in Kathmandu in the 1960s, and the standard gesture set is published by the National Federation of the Deaf Nepal [1, 2].

The community that uses it is larger than official figures suggest. The 2011 national census recorded roughly 80,000 deaf or hard-of-hearing Nepalis. The National Federation of the Deaf Nepal puts the real figure above 300,000 and attributes the difference to the undercounting of a condition that is not visible to an enumerator [3, 4]. For most of these people sign language is the primary means of communication, and almost none of the hearing population can understand it.

This gap is inconvenient in ordinary life and dangerous in an emergency. A deaf person who needs to report a fire, ask for an ambulance, or tell a bystander that they cannot breathe has no reliable way to do so if no interpreter is present, and interpreters are scarce and cannot be summoned on demand. The time spent failing to communicate is time that matters.

Recognition technology has advanced quickly for well-resourced sign languages, particularly American Sign Language, but NSL has received very little of that attention. Only one public NSL dataset exists and it covers the fingerspelling alphabet [2]. The handful of published NSL recognition systems target individual letters or static handshapes [5–7]. Nothing published addresses emergency communication, and nothing addresses whole phrases, which are what a person actually needs to sign under pressure.

Two developments make a lightweight system practical. Pretrained hand and body tracking such as MediaPipe [8] extracts the geometry of a signing person from an ordinary camera without gloves or depth sensors, and reduces each video frame to a few hundred numbers. Sequence models such as the bidirectional Long Short-Term Memory network [9, 10] classify the resulting motion directly. Together they allow a recognition system to be trained from a few hundred recordings and to run on hardware people already own.

This project applies that combination to a narrow, high-value problem: recognising six NSL emergency phrases from a live camera feed and delivering each as on-screen text and speech. The six phrases are *I can't breathe* (मलाई सास फेर्न गाह्रो भइरहेको छ), *The building is on fire* (भवनमा आगो लागेको छ), *Call the police* (प्रहरीलाई फोन गर्नुहोस्), *I need an ambulance* (मलाई एम्बुलेन्स चाहियो), *Help me / I am in danger* (मलाई मद्दत गर्नुहोस् / म खतरामा छु), and *I need to go to the toilet* (मलाई शौचालय जानु छ). They span medical, fire, crime, and basic-need situations. Their signed form was established with Deaf experts before any data was recorded, so what the system recognises is real NSL and not an invention of the authors.

1.2 Problem Statement

In an emergency, clear communication is the difference between help arriving and help not arriving. Deaf and hard-of-hearing people in Nepal are at a disadvantage in exactly those moments:

- Most hearing people, including first responders and bystanders, do not understand Nepali Sign Language.
- Human interpreters are scarce and are almost never present when an emergency begins.
- Existing sign language recognition systems are built for high-resource sign languages. No system or dataset exists for NSL emergency communication [2, 6, 7].

A deaf person in an emergency may therefore be unable to convey an urgent need to the people around them. What is missing is an accessible, low-cost tool that recognises essential NSL emergency phrases as they are signed and presents them in a form a hearing person understands immediately.

1.3 Objectives

The primary objective was to design and implement a system that recognises six Nepali Sign Language emergency phrases and outputs them as text and audio. The specific objectives were:

- To establish the natural signed form of each of the six target phrases with NSL experts, so that the recorded data reflects how the phrases are actually signed.

- To build a custom dataset of NSL emergency phrases across multiple signers and recording conditions, since no suitable public dataset exists.
- To develop a landmark-based recognition pipeline using MediaPipe Holistic, representing each frame as 225 pose and hand coordinates.
- To normalise landmarks against anatomical anchors and standardise every clip to a fixed length derived from the recorded data.
- To train a bidirectional LSTM classifier and evaluate it on signers absent from its training data.
- To deploy the trained model in an application that, on the user's command, captures a sign and displays the recognised phrase and speaks it aloud, and to withhold output when the system is not confident.

1.4 Scope of the Project

The system recognises exactly seven classes: the six emergency phrases listed above and a seventh negative class covering non-signing motion, idle movement, and gestures outside the target set. Each input is one complete phrase performed once. Recognition works from an ordinary camera with no gloves, markers, or depth sensors, and the model runs without a GPU.

The seventh class deserves a note. A classifier trained on six phrases will assign one of those six to any input it receives, including someone scratching their head. For a system whose output is an emergency announcement, that behaviour is unacceptable. The negative class gives the model explicit examples of what is not a sign, so the confidence threshold described in Chapter 3 can withhold output for inputs unlike anything in the six target phrases.

Outside the scope are sentence-level or conversational translation, the full NSL vocabulary, recognition of more than one person at a time, and facial expression. Facial landmarks were excluded after the NSL consultants confirmed that none of the six target phrases depends on facial expression to be understood. The system is a proof of concept for a fixed phrase set and is built so that phrases, signers, and deployment targets can be added later.

CHAPTER 2

LITERATURE REVIEW

Sign language recognition sits at the intersection of computer vision and sequence modelling. Published work divides along two axes: whether the input is raw pixels or extracted landmarks, and whether the signs studied are static handshapes or motion. Emergency phrases are whole dynamic signs, so this project is concerned with the landmark-based and temporal branches of the field. This chapter reviews the works closest to that problem, summarises them, states the gap they leave, and then sets out the theory the system is built on.

2.1 Review of Related Works

2.1.1 NSL Dictionary (Acharya and Sharma, 2003)

Acharya and Sharma [1] compiled the *Nepali Sign Language Dictionary* for the National Federation of the Deaf Nepal. It remains the standard reference for the NSL gesture set. Being a printed reference it offers no recognition capability, and it does not cover the multi-sign emergency phrases this project targets, which is why the signed form of each phrase had to be established directly with Deaf experts.

2.1.2 ASL Character Recognition Using CNN (Masood et al., 2018)

Masood et al. [11] trained a convolutional neural network on raw images for American Sign Language character recognition and reported strong accuracy on a large image dataset. The work shows that convolutional networks handle static fingerspelling well when data is plentiful. It also illustrates how much data a pixel-based model needs, and it does not address motion.

2.1.3 Word-Level Sign Recognition Using CNN and RNN (Masood et al., 2018)

In a companion paper, Masood et al. [12] combined a convolutional network with a recurrent network to recognise word-level gestures from video. This established the pattern of pairing spatial feature extraction with temporal modelling for dynamic signs, which is the same principle applied here through a BiLSTM. The computational cost of running a convolutional network on every frame is the reason this project extracts landmarks instead.

2.1.4 NSL Translation Using CNN (Mali et al., 2018)

Mali et al. [5] built one of the earliest NSL recognition systems, a convolutional translator over a very small vocabulary. The signer was required to wear red gloves at test time, which limits practical use, and the system handled only static gestures.

2.1.5 MediaPipe Hands (Zhang et al., 2020)

Zhang et al. [8] introduced MediaPipe Hands, a pretrained pipeline that converts each video frame into 21 three-dimensional hand keypoints in real time on ordinary hardware. By discarding background, lighting, and skin appearance it makes compact and data-efficient gesture representations possible. MediaPipe performs no classification of its own; it is the enabling technology on which this project's representation is built.

2.1.6 Deep Learning for NSL Gesture Recognition (Ligal and Baral, 2022)

Ligal and Baral [13] explored deep architectures for NSL gesture recognition, pairing a convolutional network with a recurrent network in one variant and with a Vision Transformer in another. The work confirms research interest in NSL. It follows the pixel-based formulation and does not target emergency communication or deployment on modest hardware.

2.1.7 ISL Recognition Using MediaPipe Holistic (Goyal and Velmathi, 2023)

Goyal and Velmathi [14] applied MediaPipe Holistic to Indian Sign Language and used LSTMs for dynamic signs and convolutional networks for static ones. This is the closest published work in method to the approach taken here and supports the landmark-plus-temporal-model direction. It targets Indian Sign Language, and it produces no audio output.

2.1.8 The NSL23 Dataset (Sunuwar et al., 2024)

Sunuwar et al. [2] released NSL23, the first public NSL dataset, containing 630 videos of the 49 alphabet characters performed by 14 volunteers. It is distributed as video because some NSL signs are defined by motion. NSL23 demonstrates that landmark-friendly NSL video can be collected from a modest number of signers, which is useful evidence for a project that had to record its own data. It covers the alphabet only.

2.1.9 NSL Letter Detection Using MediaPipe and CNN (Shrestha et al., 2024)

Shrestha et al. [6] present the closest prior landmark-based NSL system. MediaPipe landmarks are re-rendered as skeleton images and a convolutional network is trained on those images, with high reported accuracy. Two observations matter here. The work confirms that MediaPipe extracts usable NSL hand geometry. It also converts the numeric landmark geometry back into pixels instead of feeding the sequence to a temporal model, and its evaluation does not separate signers between training and test, so the reported accuracy does not establish how the system behaves on a person it has not seen.

2.1.10 NSL Characters Recognition with Deep Learning (Poudel et al., 2025)

Poudel et al. [7] introduced a 36-class static image dataset and reported roughly 90% accuracy with fine-tuned MobileNetV2 and ResNet50 networks. The work confirms continued activity in NSL recognition and the continued dominance of the static-image formulation, which leaves motion-based phrase signs unaddressed.

2.2 Review Matrix

Table 2.1 summarises the reviewed works against the goal of this project.

Table 2.1: Review matrix of related works

Author (Year)	**Approach / Focus**	**Limitation for this project's goal**
Acharya & Sharma (2003) [1]	Printed dictionary of the standard NSL gesture set	Reference only; no recognition; no emergency phrases
Masood et al. (2018) [11]	CNN on raw images for ASL characters	Static only; large data requirement; not NSL
Masood et al. (2018) [12]	CNN and RNN on video for word-level gestures	Compute-heavy; not NSL; no audio output
Mali et al. (2018) [5]	CNN-based NSL translator	Very small vocabulary; red gloves required; static only
Zhang et al. (2020) [8]	MediaPipe Hands real-time landmark tracking	Tracking only; performs no sign classification
Ligal & Baral (2022) [13]	CNN+RNN and CNN+ViT for NSL gestures	Pixel-based and compute-heavy; not emergency-focused
Goyal & Velmathi (2023) [14]	MediaPipe Holistic with temporal models for ISL	Targets ISL; no audio output
Sunuwar et al. (2024) [2]	NSL23: first public NSL alphabet video dataset	Alphabet only; a dataset, not a system
Shrestha et al. (2024) [6]	MediaPipe landmarks rasterised to images; CNN	Static only; not signer-independent; no temporal model
Poudel et al. (2025) [7]	Static image dataset; MobileNetV2 / ResNet50	Static only; excludes motion-based phrase signs

2.3 Research Gap

The reviewed literature leaves four gaps that together define this project:

1. No public NSL dataset or recognition system exists for emergency phrases. All published NSL work targets the fingerspelling alphabet or isolated characters.

2. Published NSL recognition is almost entirely static and pixel-based. None feeds numeric landmark sequences to a temporal model to recognise motion-based signs.

3. NSL recognition results are reported without separating signers between training and test, so published accuracies do not establish how the systems behave on an unseen person.

4. No existing NSL system produces both text and spoken output, and none addresses what should happen when the input is not a known sign at all.

2.4 Related Theories

2.4.1 Emergency Phrases as Dynamic, Multi-Sign Sequences

Fingerspelling spells a word letter by letter using held handshapes. An emergency phrase is a different object: a complete communicative act performed as a continuous movement. A phrase such as *I can't breathe* or *The building is on fire* involves both hands moving through characteristic trajectories, shifts in posture, and transitions between component signs, all spread across many frames. No single still image contains the phrase. The information is in the sequence, which is why each phrase is recorded as a short clip, converted to a sequence of per-frame vectors, and classified as a whole.

The natural NSL form of a phrase is also not a word-for-word rendering of its English translation. *I can't breathe* is not signed as three sequential signs for "I", "can't", and "breathe"; it may be one or two signs. Establishing the natural form of each phrase with an NSL expert was therefore a prerequisite to recording anything, and the expert's performance became the reference that every signer reproduced.

2.4.2 Landmark Extraction with MediaPipe Holistic

The central design decision is to represent each frame as a compact vector of body and hand landmarks instead of raw pixels. MediaPipe Holistic [8, 15] is a pretrained pipeline that tracks several components in a single frame:

- Body pose: 33 landmarks covering the upper body, each with x, y, z coordinates, giving 99 numbers.
- Left hand: 21 landmarks at the wrist and finger joints, giving 63 numbers.
- Right hand: 21 landmarks in the same layout, giving 63 numbers.
- Face: 468 landmarks, excluded from this project.

The three components that are used are concatenated in a fixed order to produce the 225-number feature vector used throughout the project:

$$\mathbf{v} = [\underbrace{v_0, \ldots, v_{98}}_{\text{pose (99)}}, \underbrace{v_{99}, \ldots, v_{161}}_{\text{left hand (63)}}, \underbrace{v_{162}, \ldots, v_{224}}_{\text{right hand (63)}}] \in \mathbb{R}^{225}.$$

When a hand or the pose is not detected in a frame, that block is filled with zeros, so every frame produces exactly 225 numbers regardless of what was visible.

2.4.2.1 Why Pose Landmarks Are Included

Several of the target phrases involve arm trajectories and upper-body posture that hand shape alone does not capture. A 21-point hand mesh would record the shape of the hand but not where the arm carried it, and a sweeping fire-reporting gesture would become difficult to separate from a distress call. Including the 33-point pose mesh gives the model the motion arc of the arms and torso.

2.4.2.2 Why Face Landmarks Are Excluded

Facial expression carries grammatical meaning in sign languages generally. Including all 468 face points would add roughly 1,400 numbers per frame and increase the feature vector more than sevenfold. Before the dataset was recorded, the NSL consultants were asked directly whether facial expression changes or adds meaning for any of the six target phrases. They confirmed it does not for this particular phrase set, so the face block was left out. This is a property of these six phrases and not a general claim about NSL.

2.4.2.3 Why Landmarks and Not Pixels

The landmark representation was chosen for three reasons. A model over 225 coordinates per frame can be learned from a few hundred clips, where a pixel-based network would need far more. Landmarks discard background, lighting, and appearance, which are the factors that cause a pixel-trained model to fail in a room it has not seen. The resulting model is small enough to train on a laptop and to run without a GPU.

2.4.3 Dual-Anchor Landmark Normalisation

Raw landmark coordinates depend on where the signer stands and how far they are from the camera. Neither changes the meaning of the phrase, so both are removed before the data reaches the model.

2.4.3.1 Hand Normalisation

Each hand's 21 points are translated so the wrist (landmark 0) is the origin, then divided by the distance from the wrist to the base of the middle finger (landmark 9). This makes the hand representation invariant to its position in the frame and to its apparent size:

$$\hat{\mathbf{h}}_i = \frac{\mathbf{h}_i - \mathbf{h}_0}{\|\mathbf{h}_9 - \mathbf{h}_0\| + \varepsilon}, \quad i = 0, \ldots, 20,$$

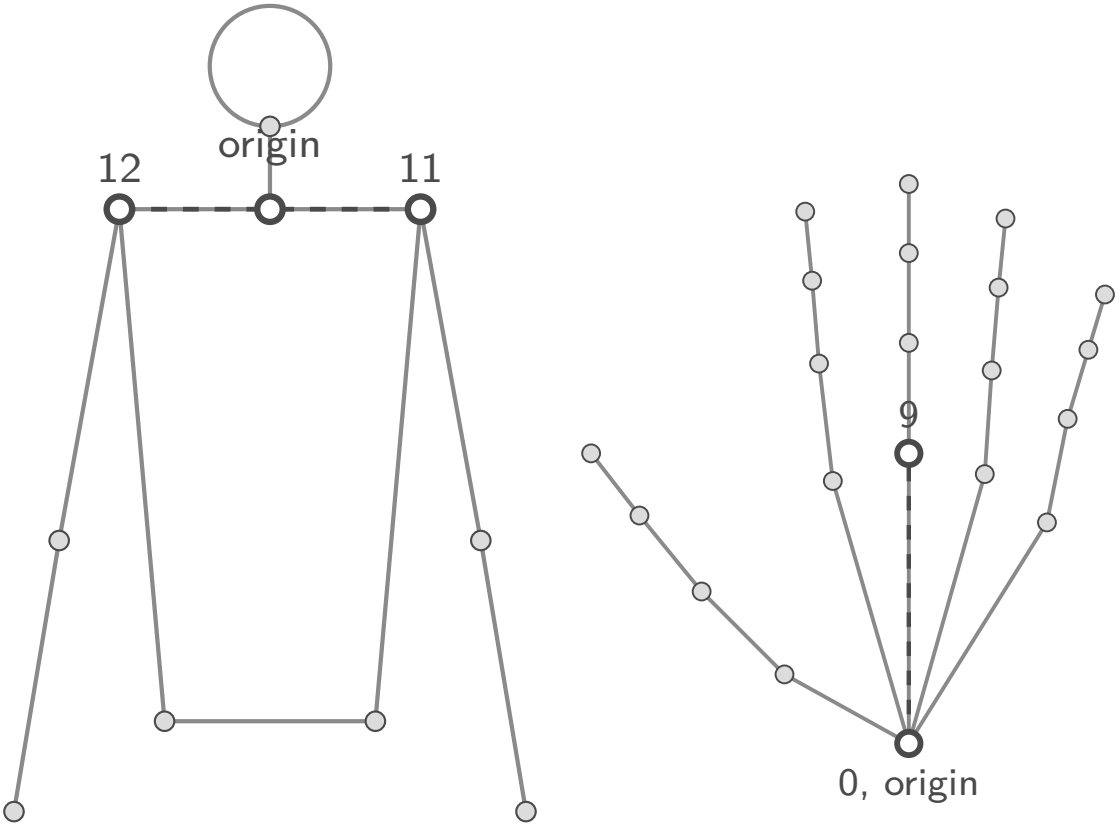
where ε is a small constant guarding against division by a near-zero scale.

2.4.3.2 Pose Normalisation

The 33 pose landmarks are translated so the midpoint of the left shoulder (landmark 11) and right shoulder (landmark 12) is the origin, then divided by the shoulder width:

$$\hat{\mathbf{p}}_i = \frac{\mathbf{p}_i - \frac{1}{2}(\mathbf{p}_{11} + \mathbf{p}_{12})}{\|\mathbf{p}_{11} - \mathbf{p}_{12}\| + \varepsilon}, \quad i = 0, \ldots, 32.$$

Figure 2.1 shows both anchors on the MediaPipe Holistic skeletons they are computed from. Landmarks 11 and 12 and their midpoint define the pose anchor; landmarks 0 and 9 define the hand anchor and the wrist-to-middle-knuckle distance used to scale it.



(a) Pose anchor: landmarks 11 and 12 (shoulders) and their midpoint

(b) Hand anchor: wrist (landmark 0) and middle-finger MCP (landmark 9)

Figure 2.1: The two normalisation anchors on the MediaPipe Holistic skeletons. Filled points are ordinary tracked landmarks; the open circles and dashed line mark the anchor points and the reference distance each block is normalised against.

Two anchors are used because one cannot do both jobs. The shoulder anchor preserves where a hand sits relative to the body, which distinguishes a sign made at the chest from one made at the side. The wrist anchor captures finger configuration independently of where the hand is. Normalising the whole 225-vector against a

single anchor would lose one or the other.

Both blocks map a zero input to a zero output, which matters later: an undetected block and a padded frame are both exactly zero and can be treated identically.

2.4.4 Sequence Standardisation

A recurrent classifier needs every input to have the same number of time steps, but signers take different amounts of time to perform the same phrase. Every clip is therefore standardised to a fixed length, seq_len. The value is measured from the recorded data instead of assumed: the frame-count distribution across all clips is computed and the 95th percentile is taken.

- A clip longer than seq_len is subsampled at seq_len evenly spaced positions across its full duration, so the whole performance is still represented.
- A clip shorter than seq_len is zero-padded at the end, never looped, with a boolean mask recording which frames are real.

The 95th percentile is preferable to the maximum because a small number of unusually long clips would otherwise force every other clip to be padded out to their length, filling the tensor with padding that carries no signal.

2.4.5 The LSTM Cell and the Bidirectional LSTM

A Long Short-Term Memory network is a recurrent network designed to carry information across many time steps without the vanishing gradients of a plain recurrent network [9]. At each step t the cell keeps a cell state c_t and a hidden state h_t, and uses a forget gate f_t, an input gate i_t, and an output gate o_t to control what is discarded, added, and exposed:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f), \tag{2.1}$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i), \tag{2.2}$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o), \tag{2.3}$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c), \tag{2.4}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{2.5}$$

$$h_t = o_t \odot \tanh(c_t), \tag{2.6}$$

where $x_t \in \mathbb{R}^{225}$ is the normalised landmark vector at step t, σ is the logistic sigmoid, $\odot$ is element-wise multiplication, and W, U, b are learned.

A standard LSTM reads in one direction, so its representation at step t depends only on the past. A bidirectional LSTM runs two such networks over the same sequence, one forward and one backward, and concatenates their hidden states [10]. This suits multi-sign phrases, where the end of a phrase often disambiguates its opening. Two phrases in this set begin with similar upward hand motion and separate only later; a forward-only model must commit to an interpretation of the opening before it has seen what follows.

2.4.6 Classification and Confidence Thresholding

The final hidden representation is passed to a fully connected layer with a softmax activation, converting raw scores z_k into a distribution over the $K = 7$ classes:

$$p_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, \qquad k = 1, \ldots, 7. \tag{2.7}$$

The predicted class is $\hat{k} = \arg\max_k p_k$ and $\max_k p_k$ is the confidence. Training minimises the categorical cross-entropy against the true label.

At inference the system emits a phrase only when the confidence exceeds a threshold and stays silent otherwise. For a tool that announces emergencies, a wrong announcement is worse than no announcement, so the asymmetry is deliberate. A softmax over seven classes is also closed-world: it distributes probability across the classes it knows and has no direct way to express that an input resembles none of them [16]. This is the reason the negative class described in Section 3.2 is part of the model itself rather than a filter bolted on afterwards: giving the classifier explicit examples of non-signing motion during training is what lets the confidence threshold do useful work against inputs that are not a sign at all.

2.4.7 Signer-Independent Evaluation

How a recognition system is evaluated determines what its accuracy means. If clips are split into training and test sets at random, clips from the same person appear on both sides. The model can then reach a high test score by recognising individuals rather than signs, and the number says nothing about a new user. This is a form of data leakage [17], and it is the weakness identified in the closest prior NSL work [6].

The alternative used here is leave-one-signer-out evaluation. Every clip is tagged with the identity of the person who recorded it. For each signer in turn, the model is trained on all clips from the other signers and tested only on that signer's clips. Every clip is scored exactly once, by a model that never saw its

signer. The mean across folds estimates accuracy on a new person, which is the quantity that matters for a tool intended for public use.

A signer can be held out only if every class is still present after their clips are removed, since a model cannot be fairly scored on a class it never saw.

CHAPTER 3

METHODOLOGY

This chapter describes how the system was built, in the order the work was carried out. The pipeline runs from establishing the signed form of each phrase with Deaf experts, through recording and preprocessing a custom dataset, to training the classifier, evaluating it, and deploying it. The evaluation protocol and the deployment path are described at the end, since both differ from what the proposal anticipated.

3.1 Software Development Approach

Agile incremental development was adopted because several parameters could not be fixed in advance. The sequence length depended on how long signers actually took to perform each phrase, the normalisation scheme had to be validated against real recordings, and the deployment target changed once the recognition results were in hand. Each increment delivered a working module: phrase verification, the recorder, the preprocessing core, the trained model, the inference server, and the mobile client.

The riskiest unknown was addressed first. Before any large-scale recording, a small set of clips was captured to confirm that MediaPipe Holistic produces stable landmarks for multi-sign NSL phrases performed at natural speed. Had it not, the entire landmark-based approach would have needed reconsideration while there was still time to change it.

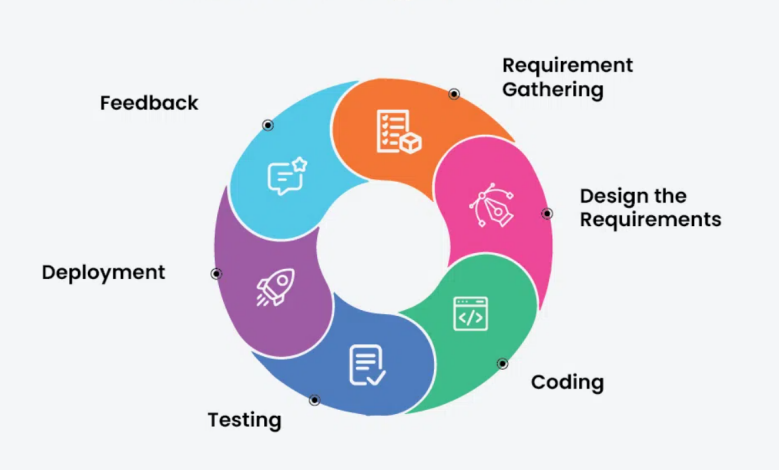


Figure 3.1: Agile incremental development approach

3.2 Phase 0: Establishing the Signed Form of Each Phrase

No standard published gloss exists for the six target emergency phrases, so the natural signed form of each had to be established with the Deaf community before recording could begin. Two consultations were held in person: with Yojana Sakha and colleagues at the Rehabilitation Centre for the Rural Deaf in Bhaktapur, and with a resource person at the National Federation of the Deaf Nepal in Ranibari Marga, Samakhushi, Kathmandu.

Each consultant was asked to demonstrate how the idea would be signed in an actual emergency, as concisely as possible, and not to sign the English sentence word by word. Each demonstration was discussed for regional variation, recorded as a reference, and confirmed as the form every signer would reproduce. Because the experts' time was limited, they also contributed roughly ten clips per phrase, which anchor the dataset to an authentic performance. Photographs from the consultations appear in Appendix A.

The consultants were also asked directly whether facial expression changes or adds meaning for any of the six phrases. They confirmed it does not for this phrase set, which settled the exclusion of the 468 face landmarks from the feature vector.

Table 3.1 lists the six phrases and the situation each covers. Phrase 6 replaced an earthquake phrase considered at proposal stage. The consultants advised that a request to use the toilet is a far more frequent communication failure for deaf people in institutional settings than an earthquake report, and the change was approved at the mid-term defence.

Table 3.1: Target emergency phrases and their categories

#	English phrase	Nepali phrase	Category	Class key
1	I can't breathe	मलाई सास फेर्न गाह्रो भइरहेको छ	Medical	`cant_breathe`
2	The building is on fire	भवनमा आगो लागेको छ	Fire	`building_on_fire`
3	Call the police	प्रहरीलाई फोन गर्नुहोस्	Crime	`call_police`
4	I need an ambulance	मलाई एम्बुलेन्स चाहियो	Medical	`need_ambulance`
5	Help me / I am in danger	मलाई मद्दत गर्नुहोस् / म खतरामा छु	Generic emergency	`help_danger`
6	I need to go to the toilet	मलाई शौचालय जानु छ	Basic need	`need_toilet`
7	Not one of the above	—	Negative class	`none`

3.2.1 The Negative Class

The seventh class was added during development and is not in the original proposal. A classifier over six phrases assigns one of those six to every input it receives, so a signer adjusting their glasses would be reported as an emergency. For a system whose output is a spoken emergency announcement, that is the worst available failure mode.

The `none` class holds deliberate non-examples: resting, idle movement, gestures outside the target set, and partial or abandoned signs. Giving the model explicit negatives lets it learn a decision boundary around the six real phrases instead of partitioning the whole input space among them, so the confidence threshold of Section 3.9.6 has a class to fall back on instead of being forced to choose one of six emergency phrases for every input.

3.3 Phase 1: Dataset Collection

Because no public dataset of NSL emergency phrases exists, one was recorded from scratch. This was the most consequential phase of the project: the ceiling on recognition accuracy is set by the quality and variety of the recordings.

3.3.1 The Recorder Application

A purpose-built recorder was written for the task. It runs MediaPipe Holistic live on a webcam feed and, for every frame, extracts a 225-value vector: 33 pose landmarks, 21 left-hand landmarks, and 21 right-hand landmarks, each with x, y, and z coordinates. The operator selects a signer identity and a phrase, then starts and stops each recording explicitly, so every clip corresponds to one deliberate performance of one phrase.

Every frame is mirrored horizontally before detection. This is recorded here because it is a constraint on everything downstream: the landmarks in the dataset describe a mirrored image, and any inference path that does not reproduce the same flip will present the model with the signer's hands swapped. Screenshots of the recorder in use appear in Appendix B.

Frames in which a hand or the pose is not detected are zero-filled for that block and the detection rate is recorded. The frame is not dropped. Each clip is therefore accompanied by a JSON sidecar holding its frame count, duration, and per-block detection rates, which made it possible to identify weak recordings without replaying them.

3.3.2 Recording Setup and Variation

Variation was introduced deliberately so the model would not learn a single recording condition:

- Lighting: bright, dim, and mixed daylight near a window.
- Background: plain wall, cluttered room, and several different rooms.
- Distance: close to the camera and at arm's length or further.
- Signers: eight people, differing in body size, hand size, and signing speed.
- Speed: each signer performed each phrase somewhat faster and somewhat slower across takes.

Each clip is saved as `<phrase>__<signer>__<index>.npy`. Encoding the signer in the filename is what makes signer-independent evaluation possible later, and it was the reason for adopting the convention.

3.3.3 Storing Landmarks Instead of Video

The recorder saves only the per-frame landmark vectors. Video is never written to disk. This keeps the dataset small, removes a slow reprocessing step from every later stage, and means no video of any signer is stored or transmitted.

The cost is worth stating plainly. Because no video was kept, the landmark representation cannot be recomputed. Changing the landmark source, for example to run detection on a phone instead of a server, cannot be evaluated against the existing data and would require re-recording every clip.

3.3.4 Aggregation Across Signers

The eight signers recorded independently on separate machines. Git was used to merge each contribution into one shared repository instead of copying files by hand, which gave a reviewable history of what each signer added and caught duplicate and misnamed clips before they entered the combined set. Screenshots of the merged repository and the resulting directory layout appear in Appendix C.

After aggregation the dataset held 840 clips: exactly 120 in each of the seven classes, contributed by eight signers. The composition is reported in full in Section 5.2.

3.4 Phase 2: Feature Extraction

Every frame of every clip is reduced to the same 225-dimensional vector, concatenated in a fixed order:

- 33 pose landmarks $\times$ $(x, y, z) = 99$ values, at indices 0 to 98.
- 21 left-hand landmarks $\times$ $(x, y, z) = 63$ values, at indices 99 to 161.
- 21 right-hand landmarks $\times$ $(x, y, z) = 63$ values, at indices 162 to 224.

A clip is therefore a $(n_{\text{frames}}, 225)$ array of 32-bit floats, with n_{frames} varying between clips. This layout, the normalisation anchors, and the sequence length are held as constants in a single module that both the training code and the inference server import, so the two cannot drift apart.

3.5 Phase 3: Determining the Sequence Length

The classifier requires a fixed number of time steps, but raw clip lengths vary because signers take different amounts of time. The value was measured from the data rather than chosen.

Across the 840 clips, frame counts ranged from 18 to 217, with a median of 89 and a mean of 90.1. The 95th percentile of this distribution gives seq_len = 137 frames. The distribution and the cutoff are shown in Figure 5.1 in Chapter 5. The value is written to a metadata file that every later stage reads, so no stage recomputes or assumes it.

An observation from the same metadata turned out to matter more than expected. The recorder never locked a frame rate; it ran as fast as the interface and MediaPipe allowed, giving a median capture rate of 15.7 frames per second across a range of 10.8 to 21.2. Since the model receives a fixed-length sequence with no explicit time axis, it reads frame count as duration. The capture rate is therefore a property of the training data that any real-time path has to reproduce, and Section 3.9.4 describes how.

3.6 Phase 4: Normalisation and Standardisation

Each clip is normalised and then standardised to seq_len, in that order.

Normalisation applies the dual-anchor scheme of Section 2.4.3 to each frame, treating the three landmark blocks independently. The pose block is re-centred on the shoulder midpoint and divided by shoulder width; each hand block is re-centred

on its wrist and divided by the wrist-to-middle-knuckle distance. Normalising per block keeps the position of a hand relative to the body while removing the signer's position and distance from the camera.

Standardisation then forces the clip to exactly 137 frames in one of three ways. A clip already at 137 frames is used unchanged. A longer clip is subsampled at 137 evenly spaced positions spanning its full duration, so the whole performance is represented instead of the first 137 frames. A shorter clip is zero-padded at the end, and a boolean mask records which frames are real.

The order is what makes the padding safe. Both normalisation formulas map a zero input to zero output, so an undetected block stays zero through normalisation, and padding added afterwards is also exactly zero. The classifier's masking layer then skips any all-zero time step at no cost. Padding after normalising would have produced non-zero padded frames that the model would read as signal.

The compiled dataset is stored as three aligned arrays: the normalised sequences with shape $(840, 137, 225)$, the integer labels with shape $(840,)$, and the padding mask with shape $(840, 137)$. A manifest records the class, signer, source file, and original frame count for every row, which keeps each compiled clip traceable to the recording it came from.

3.7 Phase 5: Model Training

3.7.1 Architecture

The classifier is a bidirectional LSTM built with TensorFlow and Keras:

Table 3.2: Classifier architecture

Layer	**Purpose**
Input $(137, 225)$	one normalised clip
Masking (mask value 0.0)	skips zero-padded and undetected frames
Bidirectional LSTM, 64 units	reads the motion in both directions
Dropout 0.3	regularisation [18]
Bidirectional LSTM, 32 units	summarises the sequence
Dropout 0.3	regularisation
Dense 32, ReLU	feature layer feeding the classification head
Dense 7, softmax	probability over the seven classes

The network has 192,007 trainable parameters, which is small enough to train on a laptop CPU in minutes. The layer sizes above are not asserted. Section 5.8 compares this architecture against six variants of it, under the evaluation protocol of Section 3.8.3.

3.7.2 Why This Model and Not a Larger Pretrained One

A dataset of 840 clips is small for a network trained from scratch, which raises the question of whether a pretrained model should carry more of the work. The answer is that one already does, and that the division of labour is where the data efficiency comes from.

MediaPipe Holistic is the pretrained component. It was trained on far more human pose and hand data than this project could ever collect, and it performs the part of the task that needs that scale: locating a body and two hands in an arbitrary image under arbitrary lighting. What reaches the BiLSTM is not an image but 225 already-interpreted coordinates, so the only thing left to learn is how those coordinates move over time for each phrase. That is a much smaller problem, and 192,007 parameters is a proportionate model for it.

A pretrained video classifier operating on pixels was considered and rejected for two reasons. It would need the raw video, which was deliberately never stored, so adopting one would mean re-recording the entire dataset. It would also be too large for the deployment target, where the whole exported model is 879 kilobytes and runs without a GPU.

The dataset was also enlarged during development in preference to adding model capacity, growing from four signers and 570 clips at the mid-term to eight signers and 840 clips. For a landmark-based classifier at this vocabulary size, additional signers buy more than additional parameters, because the quantity in short supply is variation between people and not model expressiveness.

Both halves of that judgement were tested afterwards rather than left as reasoning. Widening the recurrent layers to 128 and 64 units and the dense head to 64, which is 2.8 times the parameter count, moves leave-one-signer-out accuracy by four tenths of a percentage point, while the spread between held-out signers is several times that. The measurements are reported in Section 5.8.

3.7.3 Training Configuration

Training minimises sparse categorical cross-entropy with the Adam optimiser [19] at a batch size of 16, for at most 200 epochs. Class weights are computed to be inversely proportional to class frequency, which matters more for the folds than for the whole dataset, since the classes are balanced overall but a held-out signer's contribution is not always uniform.

Two callbacks control the schedule. Early stopping monitors validation loss with a patience of 15 epochs and restores the best weights. Its minimum-improvement threshold is set to 10^{-3} and not left at zero, because with a threshold of zero any improvement at all resets the patience counter, including a change in the fourth

decimal place. Once validation loss approaches zero it keeps setting negligible new records, so training does not end on convergence; it ends whenever an optimiser instability finally breaks the streak. That made the epoch count a record of when a random fluctuation occurred instead of a property of the fit.

Learning rate reduction on plateau addresses the cause of those fluctuations, halving the learning rate after 6 epochs without improvement, down to a floor of 10^{-5}. Its patience is kept well below the early-stopping patience so that the reduced rate has a chance to take effect before training halts.

Reproducibility required more than fixing the random seed. TensorFlow's parallel CPU reductions vary between runs, and on identical data this moved per-fold accuracy by up to 2.7 percentage points. Deterministic operations are therefore enabled explicitly, at some cost in speed, so that a reported fold accuracy is a property of the data and the recipe, not of one particular run.

3.8 Phase 6: Evaluation Methodology

3.8.1 Leave-One-Signer-Out

The headline metric is leave-one-signer-out accuracy. For each signer eligible to be held out, the model is trained from scratch on every clip from the other signers and tested only on that signer's clips. Within each fold, 15% of the training clips are set aside as a stratified validation split for early stopping, so the held-out signer influences neither the fitted weights nor the stopping point.

A signer is eligible for holdout only if removing their clips leaves every class present in the remainder. All eight signers satisfied this, giving eight folds. Two figures are reported: the mean accuracy across folds, which estimates performance on a new person, and the pooled accuracy over all 840 predictions, where each clip is scored exactly once by a model that never saw its signer.

3.8.2 The Random-Split Contrast

A stratified 70/15/15 random split is also trained and reported. It is not an additional performance measure. Its purpose is to quantify what the signer-independent protocol is protecting against: under a random split, clips from the same person appear in both training and test, and the model can score well by recognising individuals. The gap between the two protocols is the size of that effect on this dataset, and it is discussed in Section 5.6.

3.8.3 The Architecture Comparison

The architecture in Table 3.2 fixes several numbers — two recurrent widths, a dense width, a dropout rate, and the choice of LSTM over GRU — that were settled before any accuracy had been measured. A comparison across seven variants of it is therefore also run and reported. Like the random-split contrast, it is not an additional performance measure; its purpose is to establish whether those numbers were a defensible choice rather than an arbitrary one.

The seven candidates vary one aspect of the design at a time: a smaller network (32/16/16 units), a larger one (128/64/64), a single recurrent layer instead of two, the removal of the dense head, dropout raised from 0.3 to 0.5, and the LSTM cells replaced with GRU cells. The seventh candidate is the shipped architecture itself, reconstructed to the same specification, so that it is measured on equal footing with the alternatives instead of being compared against a figure obtained some other way.

Every candidate is evaluated over the same eight leave-one-signer-out folds described above, with the same seed, the same stratified validation split, the same class weights, and the same early-stopping and learning-rate schedule. Only the architecture changes, so any difference in accuracy is attributable to it and not to a different training regime. The results are given in Section 5.8.

3.8.4 Metrics

Overall accuracy is reported alongside per-class precision, recall, and F1:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1} = \frac{2\,\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{3.1}$$

A confusion matrix pooled over all folds identifies which classes are mutually confusable. For an emergency system the two directions of error are not equivalent: failing to recognise a real phrase leaves the user without help, while announcing a phrase for a non-sign produces a false alarm. Both appear in the matrix and are discussed separately.

3.9 Phase 7: Deployment

3.9.1 Revision of the Deployment Approach

The proposal specified a browser application built with Next.js, running the model in the visitor's browser through TensorFlow.js, with no server involved. The delivered system is an Android application that streams frames to an inference server.

Two separate decisions produced that change and they are worth separating.

The first is the choice of device. A phone is what a person has with them when an emergency happens; a laptop with a browser open is not. For a tool whose value depends on being available at the moment of need, the phone is the correct target, and this was decided on those grounds. The browser route was not attempted. The change was recorded as a task deviation in the project log and accepted by the supervisor before the client was built.

The second decision concerns where inference runs, and it carries a real cost. MediaPipe Holistic, the exact landmark extractor the dataset was recorded with, exists as a Python pipeline. Running recognition on the device would mean reimplementing the 225-feature extraction and the dual-anchor normalisation against a different landmark source, and a numerical difference between the two implementations does not raise an error. It produces confident wrong answers, because the model receives numbers unlike anything it was trained on. Serving from Python means the inference path imports the same preprocessing module that the reported accuracy was measured with, so the correspondence between measured and deployed behaviour is structural, not something that had to be verified.

The cost is that the delivered system requires a network connection at use time, which the proposal stated it would not. For an emergency tool this is a genuine weakness and not a neutral trade. It is listed among the limitations in Chapter 6, along with the work that would remove it.

3.9.2 Model Export

The trained model is converted to TensorFlow Lite [20] so the server can run it without a full TensorFlow installation, reducing the deployed image from about three gigabytes to one. The conversion required a workaround: on the pinned TensorFlow and Keras versions used here, the standard converter entry points abort the host process inside the compiler backend instead of raising a catchable exception. Freezing the model's variables to constants before conversion avoids the failure.

Because a silent conversion error would be indistinguishable from a model that simply performs worse, the export is verified and not assumed. The converted model and the original are run on 30 real clips and compared; the export is rejected if they disagree. The measured agreement is reported in Section 5.10.

3.9.3 Inference Server

The server is a FastAPI application exposing a WebSocket endpoint. A client opens a session, streams JPEG frames while the user signs, and sends a completion

message; the server replies with the decision. Landmark extraction costs about 38 milliseconds per frame and accounts for almost the entire pipeline cost, so streaming frames during signing overlaps that work with the signing itself. When the client signals completion, only normalisation, standardisation, and the forward pass remain, which together take about 15 milliseconds. Uploading a finished recording instead would have serialised the two and added several seconds of silence before every result.

Each session holds its own MediaPipe graph, which occupies about 150 megabytes, so concurrency is bounded by memory, not processor time. A semaphore caps simultaneous sessions and refuses further connections with an explicit error instead of allowing the process to be killed for exhausting memory. Requests are authenticated with a shared key compared using a constant-time function, so that a wrong key cannot be narrowed down by timing the response. The health endpoint is left unauthenticated so that container health checks and the client's reachability test continue to work.

3.9.4 Frame Rate as a Correctness Constraint

Because the model reads frame count as duration, a client capturing at 30 frames per second produces twice as many frames for the same sign as the training data contained, and presents the model with a temporal scale it has never seen. The server therefore discards incoming frames down to the 15.7 frames per second of the training data before running detection, using an accumulator that tracks when the next frame is due. A simpler test of whether enough time has elapsed since the last accepted frame aliases a 16 frames-per-second client down to 8. The client caps its own send rate at 16 frames per second, slightly above the target so that jitter does not starve the server below it.

3.9.5 Mobile Client

The Android client performs no recognition. It captures camera frames, encodes them, sends them, and speaks the result.

Frame encoding is the client's bottleneck, since it is pure Dart. Colour conversion, rotation, and downscaling are folded into a single pass over the output pixels, avoiding the intermediate full-resolution buffers a naive implementation allocates. Frames are downscaled to 320 pixels wide and encoded with chroma subsampling, on the grounds that landmarks are derived from luminance structure and not chroma resolution. The work runs on a persistent background isolate. When a frame is already in flight the next one is dropped instead of queued, since the server discards surplus frames anyway and a queue would only add latency.

Two details are recorded because they are easy to get wrong. The application is locked to portrait orientation and rotates each frame by the camera's sensor orientation before sending, because Android delivers frames in sensor orientation and MediaPipe detects nothing at all in an image of a person lying sideways. Frames are sent unmirrored with a flag instructing the server to apply the same horizontal flip the recorder used, and the flag is set for both front and rear cameras: the lens faces the signer either way, so the signer's right hand appears on the image-left in both cases, and making the flag depend on the camera would swap the two hand blocks in the feature vector for one of them.

3.9.6 The Decision Rule

Two outcomes are possible:

1. If the predicted class is one of the six emergency phrases and the softmax confidence reaches 0.75, the result is `accepted`. The phrase is displayed and spoken.

2. Otherwise the result is `low_confidence`. This covers a prediction of the `none` class and a low-confidence prediction of any class alike: the user is told the input was not recognised clearly, and nothing is spoken.

Folding rejection into the classifier this way keeps a wrong announcement and no announcement on the same footing: both routes into `low_confidence` withhold output rather than guessing. The best guess and the top three probabilities are returned in every case, including refusals, which is what makes a failure diagnosable during a demonstration.

CHAPTER 4

SYSTEM DESIGN

This chapter records the requirements the system was built against, the feasibility assessment made before construction, and the design of the delivered system. The diagrams describe what was built and not what was proposed; the block diagram in particular differs from the one in the proposal, for the reasons given in Section 3.9.1.

4.1 Requirement Specification

4.1.1 Software Requirements

Table 4.1: Software used, and the role of each component

Component	**Role in the system**
Python 3.11	Recording, preprocessing, training, and the inference server. The version is pinned because MediaPipe and TensorFlow resolve to a compatible set only on 3.11.
MediaPipe Holistic	Pretrained pose and hand tracking. Produces the 225 landmark values per frame, offline during recording and again on the server at inference.
OpenCV	Camera capture, frame decoding, colour conversion, and the horizontal mirror applied before detection.
TensorFlow / Keras	Building and training the BiLSTM, and exporting it.
NumPy	Landmark arrays, normalisation arithmetic, and the compiled dataset tensors.
scikit-learn	Stratified splits, class weights, and the confusion matrix and classification report.
Matplotlib	Training curves, the confusion matrix, and the frame-count histogram.
TensorFlow Lite	The deployed model format. Runs on the server through a runtime that does not require full TensorFlow.
FastAPI and Uvicorn	The inference service and its WebSocket endpoint.
Docker	Packaging the server with its model and runtime so deployment does not depend on the host environment.
Flutter and Dart	The Android client: camera capture, frame encoding, the WebSocket protocol, and speech output.
Git and GitHub	Version control for the code, and the mechanism by which eight signers' recordings were merged into one dataset.

4.1.2 Hardware Requirements

Training and evaluation ran on an ordinary laptop with no GPU. The model has 192,007 parameters and each evaluation fold completes in minutes on a processor alone, so no accelerator was required at any stage. Data collection needed only a webcam. At use time the system needs an Android phone with a camera and a network connection, and a machine to run the server container.

4.1.3 Functional Requirements

1. **Dataset recording.** The system shall record clips of each target phrase, extract 225 landmark values per frame, and store each clip with its class, signer identity, frame count, and per-block detection rates.
2. **Preprocessing.** The system shall normalise every clip against the shoulder-midpoint and wrist anchors and standardise it to the fixed sequence length, using one implementation shared by the training and inference paths.
3. **Phrase recognition.** Given a captured sequence, the system shall produce a probability distribution over the seven classes.
4. **Rejection of unknown input.** The system shall be trained with an explicit negative class covering non-signing motion, so that an input unlike any of the six target phrases has a class to be assigned to instead of being forced into one of them.
5. **Confidence thresholding.** The system shall emit a phrase only when confidence reaches the configured threshold, and shall remain silent otherwise.
6. **Text and speech output.** On an accepted recognition the system shall display the phrase on screen and speak it aloud.
7. **Live capture.** The client shall stream frames while the user signs and present the result when signing ends, without the user needing to time or trim the recording.
8. **Evaluation reporting.** The system shall report signer-independent accuracy, a pooled confusion matrix, and per-class precision, recall, and F1.

4.1.4 Non-Functional Requirements

1. **Response time.** A result shall be returned promptly enough that the interaction feels immediate, which requires overlapping landmark extraction with signing instead of performing it after capture.

2. **Silence over error.** A wrong announcement is more harmful than no announcement. The system shall prefer refusing to answer.

3. **Train and serve equivalence.** Preprocessing at inference shall be the same code as preprocessing at training time, so that measured accuracy describes the deployed system.

4. **Cost.** The system shall use only free and open-source components and shall require no GPU or paid service.

5. **Privacy.** No video of any signer shall be stored. The dataset holds landmark coordinates only.

6. **Usability under stress.** The interface shall need one action to start and one to finish, with large readable output and no configuration at use time.

7. **Diagnosability.** The client shall display what the server observes per frame, so that a failure during use can be attributed instead of guessed at.

8. **Maintainability.** The feature layout, normalisation anchors, sequence length, and thresholds shall be defined in one place and imported everywhere else.

4.2 Feasibility Assessment

4.2.1 Economic Feasibility

Every component is free and open-source. Development used laptops the team already owned, and no GPU, paid dataset, or cloud service was purchased at any point. The one recurring cost of the delivered system is running the server container, which for a demonstration is a laptop and for wider use would be a small virtual machine.

4.2.2 Technical Feasibility

The enabling technologies are mature. MediaPipe Holistic provides pretrained tracking that needs no training data of its own, and the BiLSTM is a well-understood architecture for sequence classification. Reducing each frame to 225 coordinates lowers the dimensionality enough that a few hundred clips suffice, which is what made a custom dataset viable within the project timeline. The principal risk was never the model; it was whether an authentic dataset could be collected at all, which depended on access to the Deaf community.

4.2.3 Operational Feasibility

The interaction is two taps: one to begin signing and one to end. The system needs no specialised hardware on the user's side beyond a phone, addresses a need the Deaf organisations consulted during this project confirmed is real, and can be demonstrated live. The network dependency is the main operational weakness and is recorded as such in Chapter 6.

4.3 Use Case Diagram

Figure 4.1 shows the actors and their interactions. The Deaf Signer starts a session and performs an emergency phrase. The Hearing Person is the recipient of the output, reading the displayed phrase and hearing it spoken; they do not operate the system. The Developer role covers the offline work: recording and labelling the dataset, training and evaluating the model, and deploying the server.

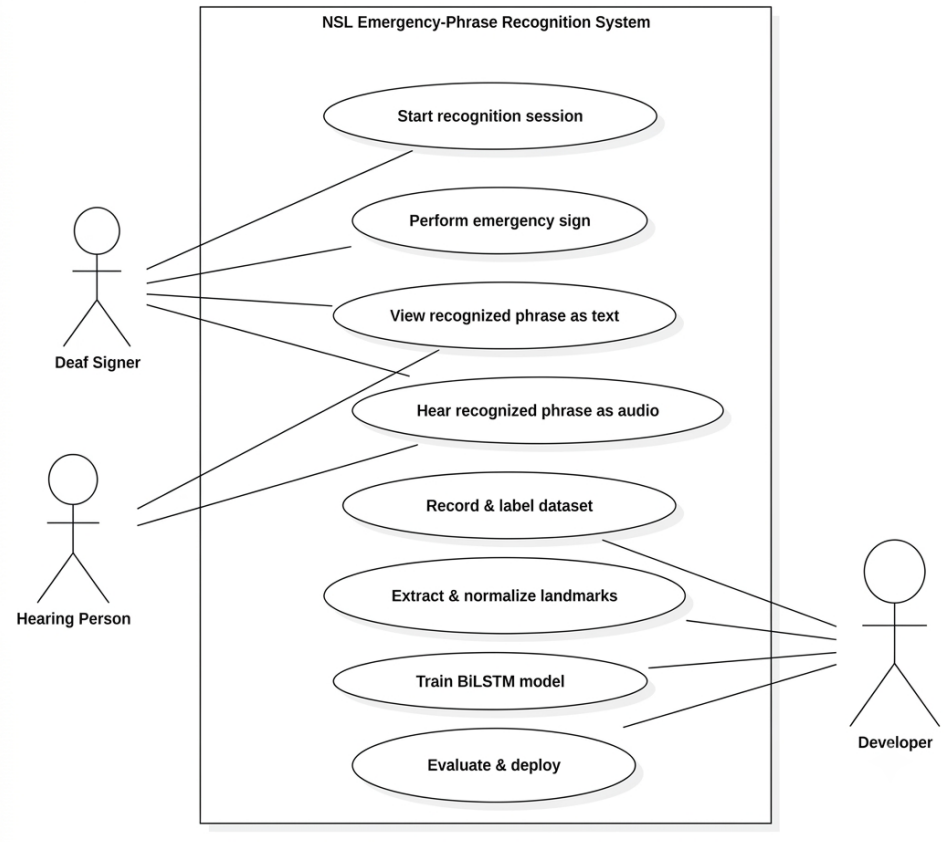


Figure 4.1: Use case diagram of the NSL emergency-phrase recognition system

4.4 System Block Diagram

The system has three parts. An offline path builds the model, a server reproduces the offline preprocessing at inference time, and a client captures the sign and presents the result. Figure 4.2 shows all three and the shared preprocessing core that connects them.

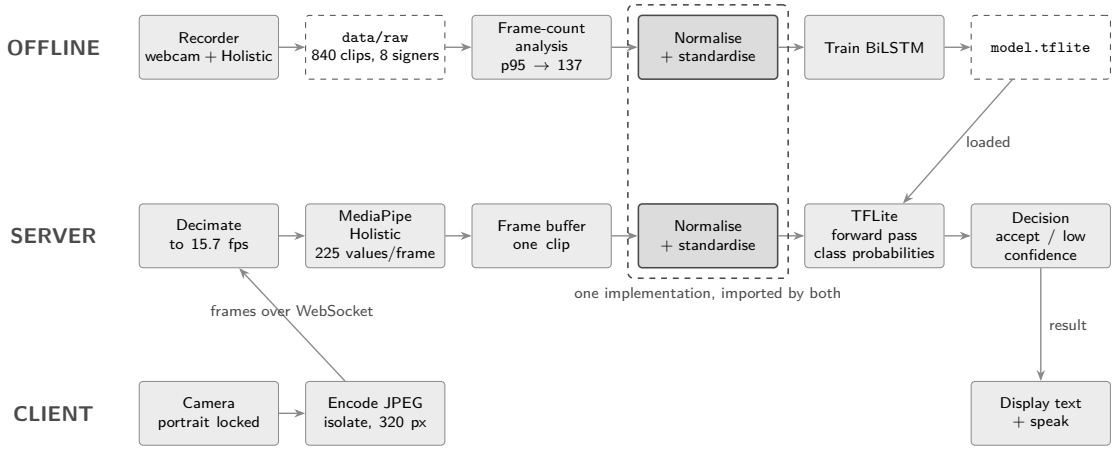


Figure 4.2: System block diagram. The offline path builds the model; the server reproduces the same preprocessing at inference; the client only captures and presents.

The shared preprocessing core is the load-bearing element of the design. It is a single module that the training scripts and the server both import, so the numbers the model sees at inference are produced by the code the accuracy was measured with.

4.5 Data Flow Diagrams

4.5.1 Level 0

Figure 4.3 shows the system as one process with its external entities. The signer supplies gestures and receives text and speech. The hearing recipient receives the same output. The developer supplies labelled recordings and the trained artefacts.

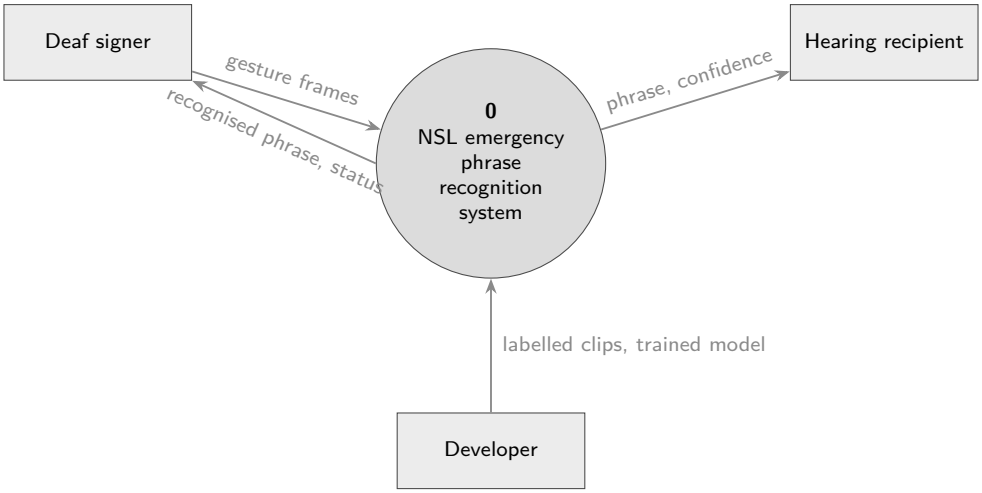


Figure 4.3: Level 0 data flow diagram

4.5.2 Level 1

Figure 4.4 decomposes the system into the processes that carry out recognition, and shows the two data stores.

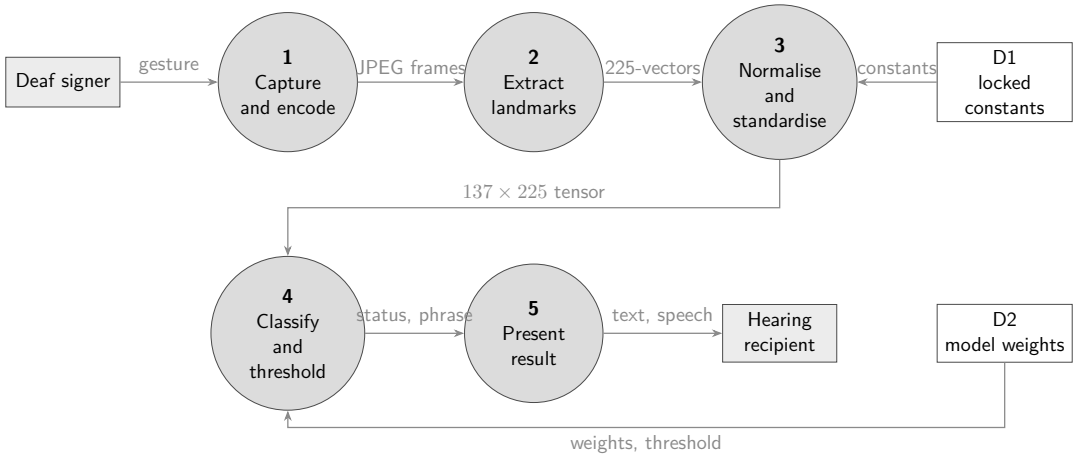


Figure 4.4: Level 1 data flow diagram

4.6 Sequence Diagram

Figure 4.5 shows one recognition from the tap that starts it to the spoken result. The loop is the important part: frames are extracted while the user is still signing, so the cost of landmark extraction is paid during the sign and not after it.

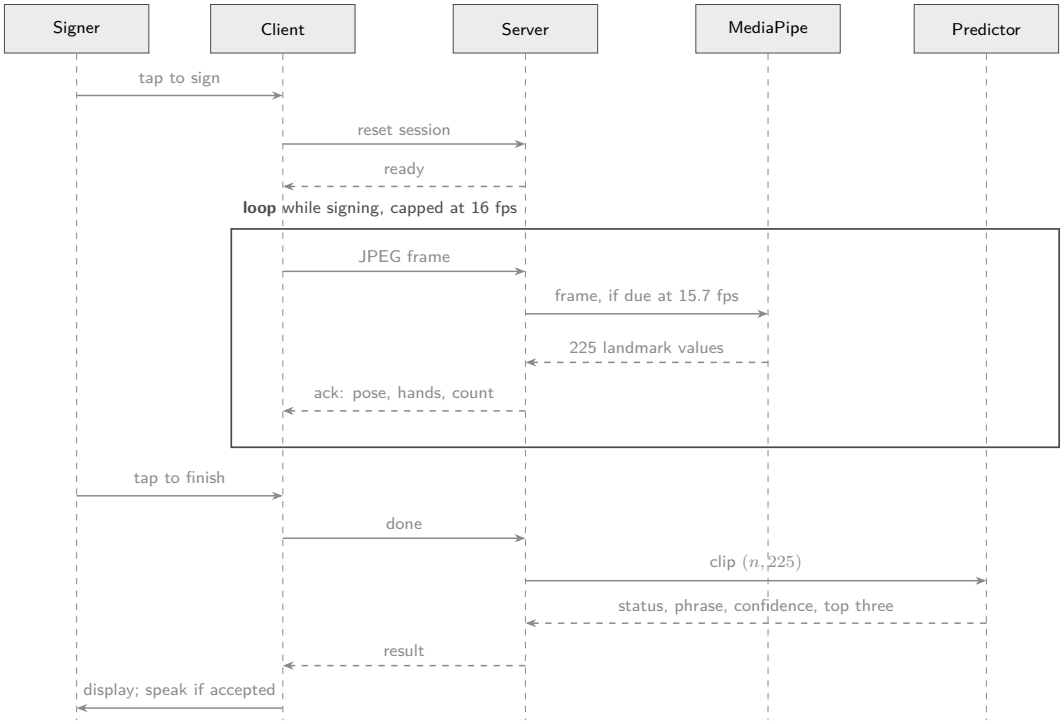


Figure 4.5: Sequence diagram for one recognition

4.7 Class Diagram

Figure 4.6 shows the principal modules and classes. The shared package on the left is imported by both the training scripts and the server, which is the structural expression of the train-and-serve equivalence requirement.

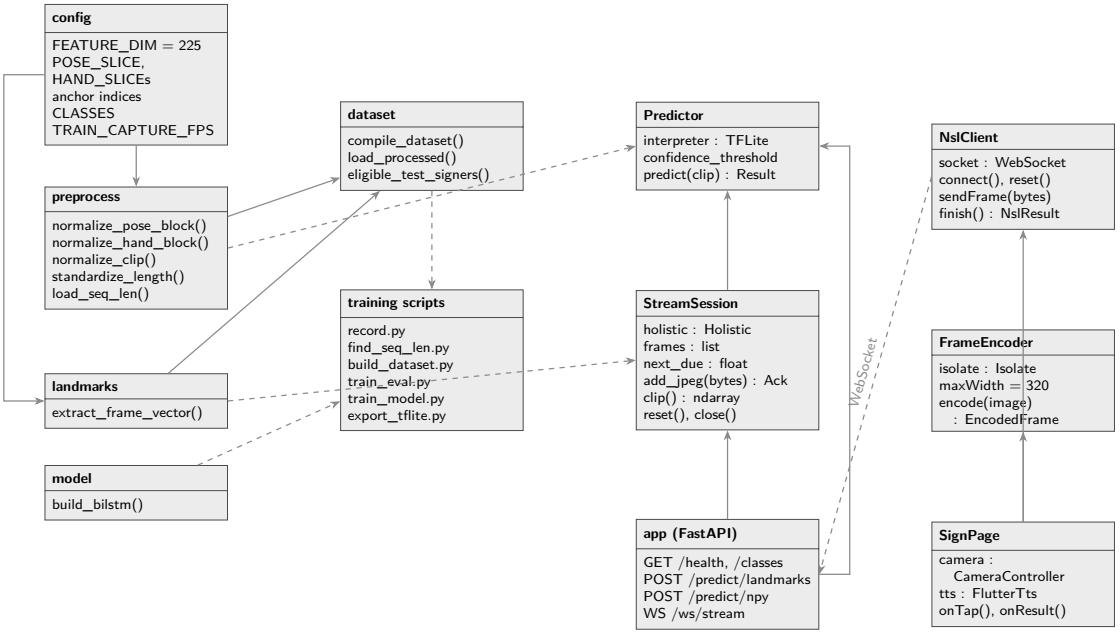


Figure 4.6: Class and module diagram. Solid arrows are use relationships; dashed arrows are imports of the shared package.

4.8 Activity Diagram

Figure 4.7 shows the control flow of one recognition attempt, including the decision point that can end it without output.

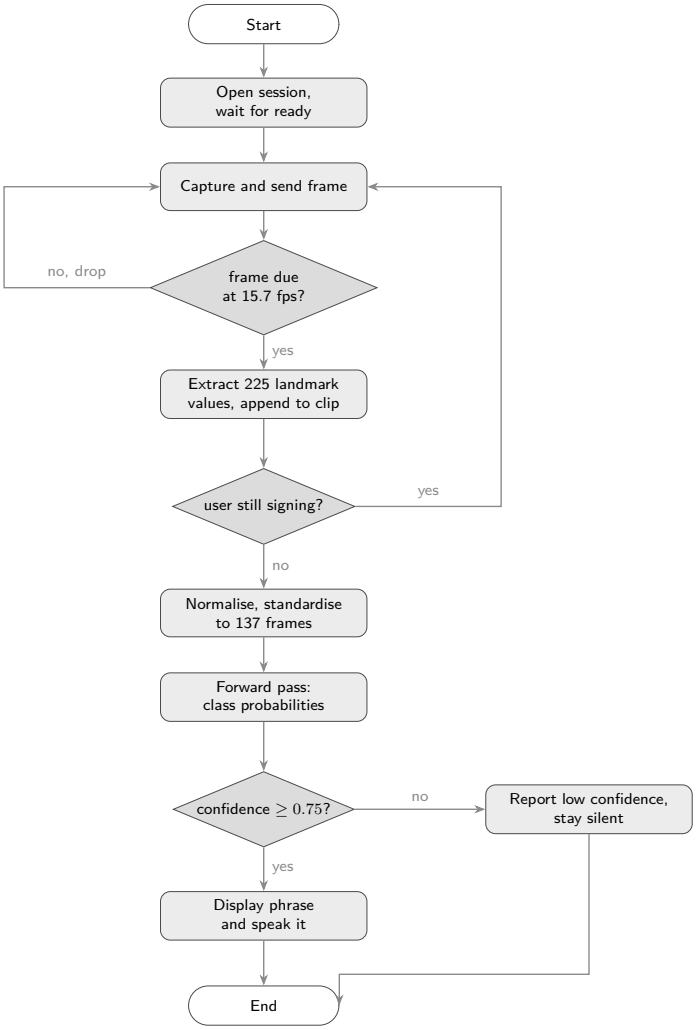


Figure 4.7: Activity diagram for one recognition attempt

CHAPTER 5
RESULT AND DISCUSSION

BEFORE SUBMISSION – DELETE THIS BOX

The four generated figures in this chapter (Figures 5.1, 5.2, 5.3 and 5.4) were produced by an earlier run and do **not** match the numbers in the tables. Figure 5.4 is titled 0.976 while Table 5.5 reports 0.9667, and Figure 5.1 was drawn from 859 clips while the text says 840. The processed tensors are also inconsistent with the recorded clips (`X.npy` is absent, and `mask.npy`, `y.npy` and `manifest.csv` hold 859 rows against 840 clips in `data/raw`), so the pipeline has to be rebuilt regardless. Run, in this order:

1. `python scripts/build_dataset.py`
2. `python -u scripts/train_eval.py --verbose 2`
3. `python -u scripts/train_model.py --verbose 2`

Then copy the regenerated `seq_len_hist.png`, `training_curves.png`, `training_loss_curves.png` and `confusion_matrix_signer_indep.png` into `img/`, and update every number in this chapter, in the abstract, and in Chapter 6 from the new `results/metrics.json` and `results/model_meta.json`.

This chapter reports what the system achieves and where it fails. The dataset is described first, then the recognition results under the signer-independent protocol, then the deployment measurements, and a discussion of the remaining weaknesses.

5.1 Experiment Setup

All training and evaluation ran on an ordinary laptop with no GPU, under Python 3.11 with TensorFlow, MediaPipe, NumPy, and scikit-learn pinned to a mutually compatible set. The classifier has 192,007 parameters, and a single evaluation fold completes in minutes on the processor alone.

The random seed was fixed at 42 for every run. Fixing the seed by itself proved insufficient: TensorFlow's parallel processor reductions vary between runs, and on identical data this moved per-fold accuracy by as much as 2.7 percentage points. Deterministic operations were therefore enabled explicitly, so the accuracies below are reproducible from the same data and code, not one sample from a distribution of runs.

5.2 Dataset Composition

The final dataset holds 840 clips, distributed exactly evenly at 120 clips in each of the seven classes. Table 5.1 gives the breakdown by class and signer.

Table 5.1: Clips by class and signer in the final dataset

Class	**s01**	**s02**	**s03**	**s04**	**s05**	**s06**	**s07**	**s08**	**Total**
building_on_fire	20	20	20	20	10	10	10	10	120
call_police	20	20	20	20	10	10	10	10	120
cant_breathe	20	20	20	20	10	10	10	10	120
help_danger	20	20	20	20	10	10	10	10	120
need_ambulance	20	20	20	20	10	10	10	10	120
need_toilet	20	20	20	20	10	10	10	10	120
none	20	20	20	20	10	10	10	10	120
Total	140	140	140	140	70	70	70	70	**840**

The four signers who recorded 20 clips per class were team members, who could record repeatedly. The four who recorded 10 per class include the Deaf consultants and additional volunteers, whose available time was shorter. Class balance was maintained deliberately, since an imbalanced set would have let the classifier gain accuracy by favouring whichever class was most common.

Eight signers is a small number, and it is the principal limitation on how much the reported accuracy can be trusted to generalise. This is discussed in Section 5.14.

5.3 Sequence Length and Standardisation

Frame counts across the 840 clips range from 18 to 217, with a median of 89 and a mean of 90.1. Figure 5.1 shows the distribution. The 95th percentile falls at 137 frames, which became the fixed sequence length.

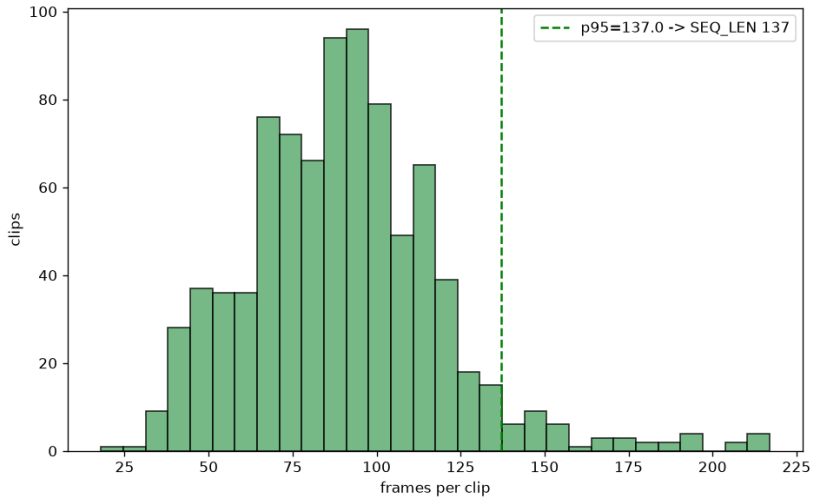


Figure 5.1: Distribution of raw clip frame counts across the 840 recorded clips, with the 95th-percentile cutoff that fixes the sequence length at 137 frames

At a capture rate of 15.7 frames per second, 137 frames corresponds to about 8.7 seconds of signing, and the median clip of 89 frames to about 5.7 seconds. Table 5.2 shows how the standardisation step treated the clips.

Table 5.2: How the 840 clips were standardised to 137 frames

Mode	**Clips**	**Treatment**
Padded ($<$ 137 frames)	797	zero-padded at the end, with a mask recording the real frames
Subsampled ($>$ 137 frames)	41	sampled at 137 evenly spaced positions
Exact (137 frames)	2	used unchanged
Total	**840**	

Of the 115,080 frame slots in the compiled tensor, 74,372 hold real recorded frames and the remaining 35.4% are zero-padding that the masking layer skips. Only 41 clips were long enough to be compressed, which matters for a reason returned to in Section 5.14: below 137 frames the model receives duration information directly in the frame count, and above it that information is normalised away.

5.4 Training Behaviour

Every fold converged and stopped on the early-stopping criterion before exhausting the 200-epoch budget. Table 5.3 reports where each fold stopped alongside its

accuracy. The spread from 24 to 67 epochs reflects how quickly validation loss stopped improving on each training subset.

Figure 5.2 shows accuracy per epoch for all eight folds and Figure 5.3 the corresponding losses. Each accuracy panel carries a dashed line at that fold's held-out score. The gap between the training and validation curves, both drawn from signers inside the training set, and that dashed line is the cost of meeting a signer the model has never seen. Presenting all eight folds shows whether the folds behaved consistently, which a single representative curve cannot.

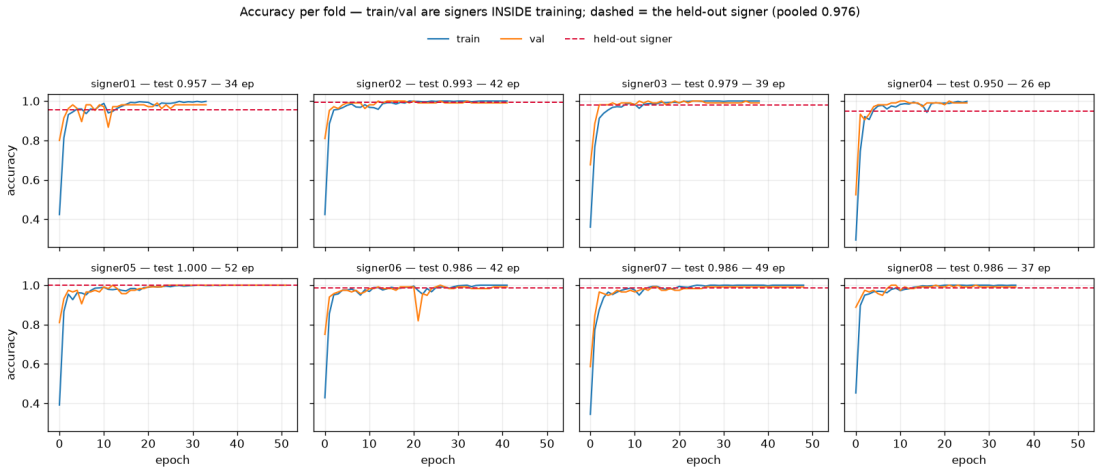


Figure 5.2: Accuracy per epoch for each leave-one-signer-out fold. Training and validation curves are drawn from signers inside the training set; the dashed line marks the held-out signer's score.

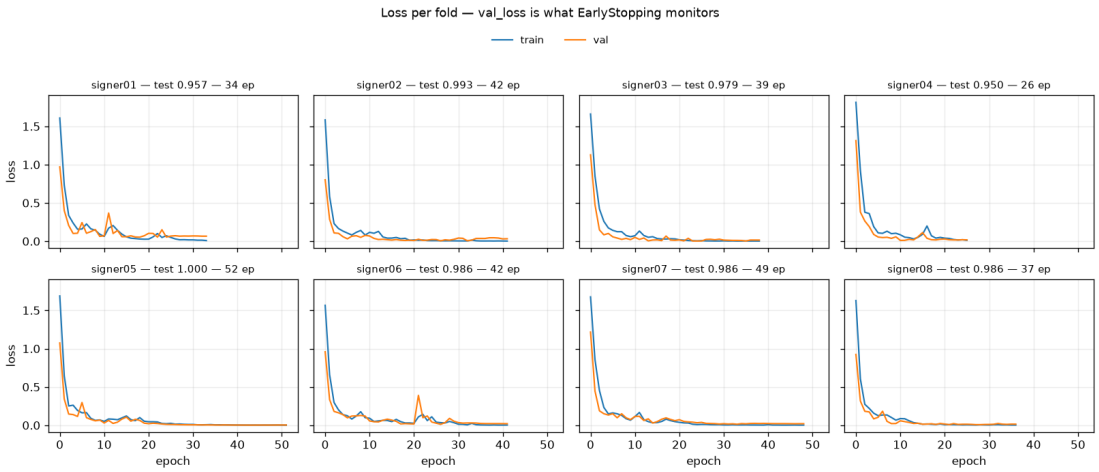


Figure 5.3: Loss per epoch for each fold. Validation loss is the quantity early stopping monitors.

5.5 Signer-Independent Recognition Results

All eight signers were eligible for holdout, since removing any one of them left every class present in the remainder. Table 5.3 gives the result of each fold.

Table 5.3: Leave-one-signer-out results. Each fold trains on seven signers and tests on the eighth.

Held-out signer	**Test clips**	**Accuracy**	**Epochs to stop**
signer01	140	0.9714	39
signer02	140	0.9786	42
signer03	140	0.9714	28
signer04	140	0.9571	44
signer05	70	1.0000	67
signer06	70	0.9857	24
signer07	70	0.9286	24
signer08	70	0.9286	49
Mean across folds		**0.9652** $\pm$ 0.0241	
Pooled over 840 clips		**0.9667**	

Mean accuracy across folds is 96.52% with a standard deviation of 2.41 percentage points. Pooling all 840 predictions, each made by a model that had not seen its signer, gives 96.67%.

The spread between folds is more informative than the mean. The best fold is perfect and the two worst sit at 92.86%, a range of about seven points. Since each fold differs only in which signer was withheld, that spread is a direct measurement of how much recognition quality depends on the individual. Signers 07 and 08 are the two hardest, and both contributed 70 clips against the others' 140, so their folds are also the ones where a single misclassification moves the score furthest. Reporting the mean alone would hide this; a system deployed to the public will meet signers at both ends of that range.

5.5.1 Per-Class Performance

Table 5.4 gives precision, recall, and F1 for each class, pooled over all folds.

Table 5.4: Per-class performance, pooled over all eight folds

Class	Precision	Recall	F1	Support
`building_on_fire`	0.9587	0.9667	0.9627	120
`call_police`	0.9667	0.9667	0.9667	120
`cant_breathe`	0.9597	0.9917	0.9754	120
`help_danger`	1.0000	0.9917	0.9958	120
`need_ambulance`	0.9748	0.9667	0.9707	120
`need_toilet`	0.9835	0.9917	0.9876	120
`none`	0.9224	0.8917	0.9068	120
Macro average	0.9665	0.9667	0.9665	840

The six emergency phrases achieve F1 between 0.963 and 0.996. The negative class is clearly the weakest at 0.907, with a recall of 0.892.

That asymmetry is expected. The six phrases are each a specific, repeatable performance, and the model has 120 examples of each. The `none` class is not a phrase; it is everything else, sampled 120 times from an unbounded space of possible non-signs. No number of examples makes that category as learnable as the six, and its recall would be expected to remain the lowest even with far more data.

5.5.2 Confusion Matrix

Figure 5.4 shows the pooled confusion matrix and Table 5.5 the same counts numerically.

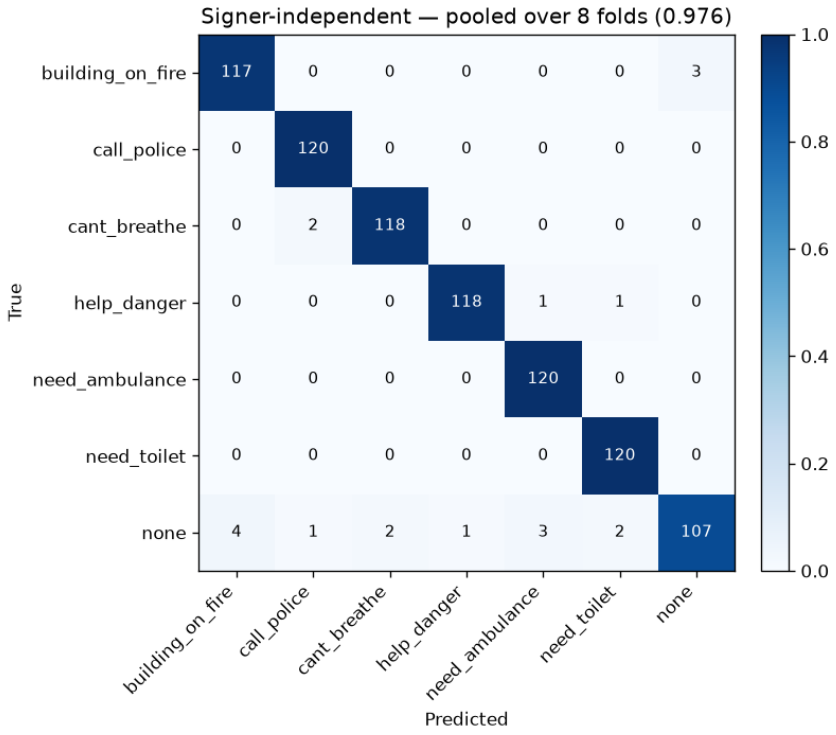


Figure 5.4: Confusion matrix pooled over all eight leave-one-signer-out folds

Table 5.5: Pooled confusion matrix. Rows are the true class, columns the prediction.

	fire	police	breathe	help	ambulance	toilet	none
building_on_fire	**116**	0	0	0	0	0	4
call_police	0	**116**	4	0	0	0	0
cant_breathe	1	0	**119**	0	0	0	0
help_danger	0	1	0	**119**	0	0	0
need_ambulance	0	0	0	0	**116**	0	4
need_toilet	0	0	0	0	0	**119**	1
none	4	3	1	0	3	2	**107**

Twenty-eight of the 840 predictions are wrong, and they fall into three groups that carry very different consequences.

Thirteen errors are non-signs predicted as emergency phrases, in the bottom row. These are the most serious errors the system can make, because each one is an occasion where the system would announce an emergency that nobody signed. They are spread across five of the six phrases, which suggests the cause is a genuine

resemblance between some non-signing motion and the opening of a phrase and not one specific confusable pair.

Nine errors are emergency phrases predicted as `none`, in the last column: four for `building_on_fire`, four for `need_ambulance`, and one for `need_toilet`. Here the system stays silent when the user did sign. This leaves the user without help, though it does not produce a false alarm, and in a real deployment the user would notice the absence of a response and sign again.

Six errors are one emergency phrase confused with another. Four are `call_police` read as `cant_breathe`, which is the only confusable pair with more than one occurrence, and the remaining two are isolated. A wrong phrase is worse than silence, because it sends the wrong help, and this is the category most worth reducing.

Notably, `help_danger` achieves perfect precision: nothing else in the dataset was ever mistaken for it. For the most generic distress phrase in the set, that is a useful property.

5.6 What the Random Split Shows

Under a stratified 70/15/15 random split, the same architecture and the same training recipe reach 99.21% accuracy on 126 held-out clips. This is 2.5 percentage points above the signer-independent figure.

That gap is the point of reporting the number. Under a random split, clips from the same person appear in both training and test, so the model can score well by recognising the individual as much as the sign. The signer-independent protocol removes that possibility, and the accuracy falls. The 99.21% figure describes how well the system recognises phrases performed by people it has already been trained on, which is not the situation any real user is in.

Table 5.6: The two evaluation protocols compared

Protocol	**Accuracy**	**What it measures**
Random split, stratified 70/15/15	0.9921	performance on known signers
Signer-independent, mean of 8 folds	0.9652	performance on a new signer
Signer-independent, pooled	0.9667	performance on a new signer

This is worth stating because published NSL recognition results are generally reported without separating signers, including the closest prior landmark-based work [6]. A comparison of headline accuracies across papers is therefore not meaningful unless the protocols match. On this dataset the difference between the two protocols is 2.5 points, and a project reporting only the higher number would not be doing anything unusual by the standards of the field.

5.7 Generalisation and Overfitting

A model with 192,007 parameters trained on 840 clips has enough capacity to memorise its training set, so whether it has done so is a question the evaluation has to answer rather than assume. Three accuracies are available for each fold, and the distance between them is what the question turns on.

Table 5.7: The three levels of accuracy, and what the gap between each pair means

Measured on	**Drawn from**	**A large gap to the next level means**
Training clips	signers inside training	the model is fitting its own examples
Validation clips	signers inside training, clips held out	memorisation of individual clips
Held-out signer	a person absent from training	memorisation of the people, not the signs

Four pieces of evidence bear on overfitting.

First, no fold ran to the epoch limit. Early stopping halted every fold between 24 and 67 epochs against a budget of 200, monitoring validation loss with the best weights restored. A model still improving on held-out data at the point it stops has not begun to memorise; a model that needed the full 200 epochs and kept its final weights would be the worrying case.

Second, the regularisation is not nominal. Two dropout layers at 0.3 sit between the recurrent layers, the learning rate halves after six epochs without improvement, and the masking layer removes padded frames from the computation so they cannot be fitted as signal.

Third, the per-fold spread is narrow. Eight independently trained models, each missing a different signer, produce accuracies between 0.9286 and 1.0000 with a standard deviation of 2.41 percentage points. A model that had memorised its training signers would be expected to collapse on some folds and not others, giving a much wider spread than this.

Fourth, and least comfortably, the random-split comparison puts a number on the memorisation that does remain. The 2.5-point gap between 99.21% and 96.67% is the portion of a random-split score that comes from recognising people rather than phrases. It is small, but it is not zero, and it is the reason the signer-independent figure is the one quoted throughout this report.

Taken together these say the model generalises to a new signer with a loss of roughly two and a half points against a same-signer measurement, and that it is not overfitting in the sense of having memorised its training clips. What they do

not say is that it generalises to a new signer drawn from outside the population of eight, which is a separate claim and is addressed in Section 5.14.

5.8 Architecture Comparison

The layer sizes in Table 3.2 were settled before any of the accuracies above existed, and nothing reported so far establishes that they were the right choice. The protocol in Section 3.8.3 tests them: seven variants of the architecture, each evaluated over the same eight leave-one-signer-out folds with the same seed, split, and callbacks, so that a difference between them is a property of the architecture and not of the training regime. Table 5.8 gives the result and Figure 5.5 shows the same numbers as intervals.

Table 5.8: Architecture variants under the leave-one-signer-out protocol, ranked by mean accuracy. Units are given as first recurrent layer / second recurrent layer / dense head; a dash marks a layer the variant does not have.

Architecture	Units	Cell	Dropout	Params	Mean	95% CI	Epochs
gru	64/32/32	GRU	0.3	145,159	0.972	[0.962, 0.982]	36
bigger	128/64/64	LSTM	0.3	535,559	0.965	[0.947, 0.983]	43
current	64/32/32	LSTM	0.3	192,007	0.961	[0.944, 0.978]	43
single_layer	64/–/32	LSTM	0.3	152,839	0.960	[0.939, 0.980]	43
no_dense_head	64/32/–	LSTM	0.3	190,151	0.952	[0.930, 0.974]	40
high_dropout	64/32/32	LSTM	0.5	192,007	0.947	[0.925, 0.970]	51
smaller	32/16/16	LSTM	0.3	77,063	0.944	[0.911, 0.976]	51

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The `current` row of Table 5.8 is the shipped architecture measured through this same sweep, and it reproduces the leave-one-signer-out result exactly (mean 0.9611, standard deviation 0.0244, matching `results/metrics.json` to every decimal place). Table 5.3 and the surrounding text still carry 0.9652 and 0.9667 from the earlier run flagged at the start of this chapter. Once those are regenerated the two tables will agree; until then they differ by 0.4 points and an examiner comparing them will notice. Regenerate Chapter 5 from the current `metrics.json` first, then delete this box.

No candidate is distinguishable from any other. The means span 0.944 to 0.972, but with only eight folds the standard error on each one is between 0.5 and 1.7 percentage points, and every confidence interval in the table overlaps every other. Even the widest separation in the set, between the GRU variant and the smallest one, is 2.8 points against a standard error of 1.7 on the difference.

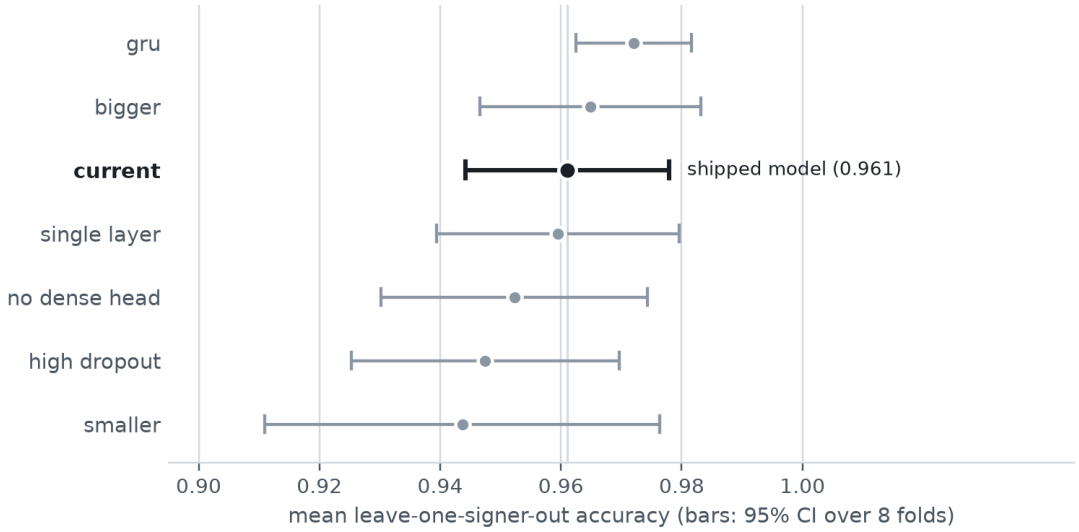

Figure 5.5: Mean leave-one-signer-out accuracy for each architecture variant, with 95% confidence intervals over the eight folds. The vertical rule marks the shipped model's mean; every interval crosses it.

The comparison that matters more is against the spread within each candidate. The fold-to-fold standard deviation of a single architecture ranges from 0.014 to 0.047, which is larger than every difference between architectures in the table. Which signer is held out affects the result more than which architecture is trained. That is the substantive finding of this section, and Section 5.14 returns to it.

Three conclusions do follow from the table. Capacity is not the bottleneck: the largest variant carries 535,559 parameters, 2.8 times the shipped model, and gains four tenths of a point for them. A dropout rate of 0.3 is the right setting, since raising it to 0.5 is nominally worse and needs 51 epochs to converge instead of 43. And the smallest variant is the only one that looks genuinely handicapped — not for its mean, which is within noise of the rest, but for a fold-to-fold standard deviation of 0.047, more than three times the GRU variant's, and the slowest convergence in the set. A network of 32 and 16 units is too small to behave consistently from one held-out signer to the next.

Two properties of the comparison limit what can be read from it. Each candidate was trained once per fold from a single seed, so run-to-run variation from weight initialisation is not measured and could account for differences of this size by itself. And eight folds cannot resolve differences of one or two points whatever statistical test is applied to them; that is a property of the design and not of the analysis.

The shipped architecture is therefore retained — not because it measurably outperforms the alternatives, but because no alternative measurably outperforms

it, and it sits at the smaller end of the range over which accuracy has already saturated. The GRU variant is the one worth revisiting should the deployment target ever impose a size or latency budget this model misses: its accuracy advantage is within noise, but its lower fold-to-fold variance, faster convergence, and 24% smaller parameter count do not depend on that difference being real.

5.9 Why the Accuracy Is High

An accuracy near 97% on a first attempt at an under-resourced sign language invites suspicion, and it should. Five properties of the task account for it, and three of them are forms of optimism that a reader should discount.

The legitimate reasons come first. The problem is a seven-way choice, and the six phrases were drawn from different semantic domains: medical, fire, crime, and basic need. Nothing constrained them to be visually distinct, but choosing by situation produced a set whose signed forms differ in handshape, location, and movement all at once. The confusion matrix reflects this directly, with one materially confusable pair out of twenty-one possible pairs. Second, the landmark representation discards background, lighting, and appearance before the model sees anything, and dual-anchor normalisation removes position and scale. The nuisance variation that costs a pixel-based model most of its accuracy is not present in the input. Third, the classes are exactly balanced at 120 clips each, so no part of the score comes from favouring a common class.

The three optimistic properties matter more for interpreting the number.

Every signer reproduced a single fixed reference form, established with the consultants before recording. This was the right methodological choice, since it fixes the ground truth for each class, but it also means the dataset contains eight people imitating one performance rather than eight people signing naturally. Real signing carries regional and personal variation that this dataset largely excludes, so intra-class variance here is lower than a deployed system would meet.

Each input is one complete phrase, performed deliberately, with the user marking its start and end by tapping. The system never has to find a phrase inside continuous motion, decide where one sign ends and the next begins, or handle a signer who changes their mind mid-phrase. Segmentation is one of the harder problems in sign recognition and this design avoids it entirely.

Half the signers are the authors, who knew the phrase set better than anyone and had the most practice performing it. Their four folds average 0.9696 against 0.9607 for the other four, a difference too small to be conclusive at this sample size but pointing the direction one would expect.

None of this makes 96.67% wrong. It makes it an accuracy on this task as specified: seven distinct classes, one deliberate phrase at a time, performed against a fixed reference by eight people, four of whom built the system. A wider vocabulary, natural signing, or continuous input would each lower it.

5.10 Model Export Verification

The trained model was converted to TensorFlow Lite for deployment, reducing it from 2.3 megabytes to 879 kilobytes and removing the need for a full TensorFlow installation on the server.

A conversion fault would be indistinguishable from a model that simply performs slightly worse, so the export is verified against the original instead of trusted. Both models were run on 30 real clips and their outputs compared. The largest absolute difference in any predicted probability was 1.2×10^{-7}, and the two models agreed on the predicted class for all 30 clips. The export step refuses to write its output if this check fails.

Table 5.9: TensorFlow Lite export verification

Quantity	**Value**
Keras model size	2.3 MB
TensorFlow Lite model size	879 KB
Clips compared	30
Largest absolute probability difference	1.2×10^{-7}
Predicted-class agreement	30 of 30

5.11 Deployment

The delivered system runs as three pieces: a container holding the server and the exported model, an Android client, and the network between them. The container omits TensorFlow entirely and runs the exported model through a lightweight runtime, which reduces the image from about three gigabytes to one.

The cost profile of a single recognition is dominated by one operation. Landmark extraction takes about 38 milliseconds per frame and accounts for almost the whole pipeline. Because frames are streamed while the user is still signing, that cost is paid concurrently with the sign, and when the client signals completion only normalisation, standardisation, and the forward pass remain, which together take about 15 milliseconds. A design that uploaded a finished recording would have serialised the two, adding several seconds of silence before every result.

Table 5.10: Measured costs on the server path

Stage	**Cost**
Landmark extraction, per frame	$\approx$ 38 ms
Normalisation, standardisation, forward pass, per clip	$\approx$ 15 ms
Frames retained after decimation	15.7 per second
Client send rate cap	16 frames per second
Memory per concurrent session	$\approx$ 150 MB
Concurrent session limit	4

Concurrency is bounded by memory, not processor time, since each session holds its own MediaPipe graph. The session limit sheds load with an explicit error instead of allowing the process to be terminated for exhausting memory.

The Android client was built and installed on a physical device, and recognition works end to end: the user taps to begin, performs a phrase, taps to finish, and the recognised phrase appears and is spoken. End-to-end latency measured from the phone, sustained client frame rate, and per-frame encoding cost are reported in Table 5.11.

Table 5.11: Client-side measurements on a physical device

Quantity	**Value**
Test device	[**device model**]
Sustained send rate while signing	[**fps**]
Per-frame encoding cost	[**ms**]
End-to-end latency, tap to spoken output	[**seconds**]

Screenshots of the client in each of its three result states appear in Appendix E.

5.12 Development Process

The work proceeded in seven increments, each delivering a module that could be exercised before the next began. Table 5.12 records what each produced and what it changed about the plan, since several increments revised decisions made earlier.

Table 5.12: Development increments

No.	Increment	Outcome and revisions to the plan
1	Consultation and phrase set	Signed form of each phrase fixed with Deaf experts. Face landmarks excluded after the consultants confirmed no phrase depends on expression. Phrase 6 changed to the toilet request.
2	Recorder and pilot clips	Confirmed MediaPipe produces stable landmarks for multi-sign phrases. Established the zero-fill convention and the signer-encoded filename that made later signer-independent evaluation possible.
3	Dataset collection and merge	840 clips across 8 signers, merged through Git. The `none` class was added here, after it became clear a six-way classifier would name a phrase for any input.
4	Preprocessing core	Sequence length derived at 137 from the measured distribution. Established that normalisation must precede padding so padded frames stay exactly zero.
5	Training and evaluation	Leave-one-signer-out results. Two corrections came out of this stage: the early-stopping threshold had to be raised off zero, and deterministic operations had to be enabled before fold accuracies were reproducible. The architecture comparison of Section 5.8 was run here, once the evaluation harness existed to run it through.
6	Confidence thresholding	Decision rule fixed on the softmax over all seven classes, accepting a phrase only above threshold. The confidence threshold was lowered from 0.85 to 0.75 once the negative class was in place, since 0.85 was rejecting genuine signs.
7	Export and deployment	TensorFlow Lite export with verification, inference server, and Android client. The deployment target changed from a browser to a phone at this stage, for the reasons in Section 3.9.1.

5.13 Behaviour as the Vocabulary Grows

The system recognises seven classes. Any extension of this work adds more, so it is worth setting out what the design predicts will happen, and which parts of it break first. No experiment in this report tests this; what follows is an argument from the structure of the method, and it identifies where the next measurements should be taken.

5.13.1 Confusion Grows Faster Than the Vocabulary

A K-class problem has $K(K-1)/2$ pairs that can be confused. At seven classes there are 21, and the confusion matrix shows one of them carrying more than a single error. The count grows quadratically: 105 pairs at fifteen classes, 435 at thirty.

The more important effect is not the arithmetic but which pairs get added. The six phrases here were chosen by situation, not by appearance, and they happened to differ in handshape, location, and movement simultaneously. A vocabulary of thirty NSL phrases cannot be chosen that way. It must include pairs that differ in one parameter only, which is what sign-language phonology calls a minimal pair, and those are exactly the cases a 137-frame landmark sequence at 15.7 frames per second is least able to separate. Two signs distinguished by a brief change in handshape occupy a handful of frames out of 137.

The exclusion of facial landmarks becomes a limitation at this point rather than a simplification. The consultants confirmed that none of the six current phrases depends on facial expression, and that confirmation does not extend to a larger set. NSL, like other sign languages, carries grammatical information on the face, including negation and question marking. A vocabulary that includes a phrase and its negation cannot be separated by the current 225-value feature vector at all, because the information that distinguishes them is discarded before the model sees the input.

5.13.2 The Data Requirement Scales With Classes, Not Total Clips

What governs recognition quality is clips per class, not clips overall. Holding the present 120 per class, a thirty-class vocabulary needs 3,600 clips, and each new class needs its own consultation to establish its signed form before recording can begin. The consultation step, not the recording, is the binding constraint: it requires expert time from a small community.

5.13.3 The Negative Class Gets Harder

The `none` class is already the weakest at 0.907 F1, and it degrades as the vocabulary grows for a structural reason. Its job is to cover everything that is not one of the known phrases. As known phrases multiply, the boundary between the known region and everything else becomes longer and more intricate, while the number of `none` clips available to describe it stays fixed. A negative class of 120 clips describing the complement of six phrases is thin; describing the complement of thirty is thinner.

5.13.4 The Confidence Threshold Needs Recalibrating

The threshold of 0.75 requires the model to place three quarters of its probability mass on one class. Softmax spreads probability across whatever the model finds plausible, so as near-neighbours multiply the maximum probability falls even when the top prediction is correct. A fixed 0.75 would therefore reject an increasing share of correct recognitions as `low_confidence`. The threshold is a function of vocabulary size and would have to be recalibrated on validation data at each vocabulary, not carried forward.

5.13.5 Predicted Shape of the Degradation

Table 5.13 collects these effects.

Table 5.13: Predicted effect of a larger vocabulary on each component

Component	**Effect of raising K**	**What would be needed**
Phrase classifier	Confusable pairs grow as K^2; minimal pairs enter the set	Higher frame rate, or explicit handshape features
Feature vector	Face-dependent distinctions become unrepresentable	Add the face block, at roughly sevenfold the input width
Training data	120 clips per new class, plus one consultation each	Expert time, the scarcest input
`none` class	Describes a larger complement with the same clip count	Negative clips scaled with K
Confidence threshold	Correct predictions fall below a fixed 0.75	Recalibration per vocabulary size

The expected trajectory is therefore not a steady decline. Adding phrases that are semantically and visually distinct from the existing six should cost little, because the representation already separates such phrases well. Accuracy should fall sharply at the point where the vocabulary first admits a minimal pair, and again where it admits a face-dependent distinction, since the second of those is not a matter of degree but of information the input does not contain.

The cheapest way to test this is to add three or four phrases chosen to be deliberately similar to existing ones and re-run the leave-one-signer-out evaluation. That measures the first of the two cliffs directly, at a cost of one consultation and a few hundred clips.

5.14 Discussion

5.14.1 What the Results Support

Under a protocol that tests only on unseen signers, the system recognises the six emergency phrases with a pooled accuracy of 96.67% and per-class F1 between 0.963 and 0.996. The landmark representation is what makes this achievable from 840 clips: a pixel-based model on the same data would have far more parameters to fit and far less invariance to the rooms and lighting the clips were recorded in.

The evaluation protocol is the part of this work most worth carrying forward. The 2.5-point gap between the random split and the signer-independent folds is a concrete measurement of how much a random split flatters a model on this kind of data, and it was obtained on a dataset small enough that the effect could easily have been dismissed as noise.

5.14.2 Eight Signers Is the Binding Limitation

Every accuracy in this chapter rests on eight people, four of whom are the authors. The standard deviation across folds is 2.41 points and the range is seven, which means the estimate of performance on a new signer is itself uncertain. A ninth signer with an unusual signing style could plausibly score below any fold reported here.

The architecture comparison in Section 5.8 puts this on a firmer footing than an assertion. Across seven architectures spanning 77,063 to 535,559 parameters, no variant is distinguishable from any other, and the fold-to-fold spread within each one is larger than every difference between them. What limits this system is not the model that was chosen but the number of people it has been measured on.

Four of the eight signers are also not fluent NSL users. They learned each phrase from the consultants' reference and reproduced it. The dataset therefore contains a mixture of fluent and learned performances, and the accuracy above is an accuracy on that mixture, not on fluent signing.

5.14.3 The Negative Class Is the Weak Point

The `none` class carries a recall of 0.892, and thirteen non-signs were predicted as emergency phrases. In deployment these are the errors that produce a false announcement, which is the most serious failure mode this system can have. All thirteen also passed the confidence threshold, meaning the classifier was confident about a wrong answer rather than merely uncertain, so raising the threshold would not by itself have caught them.

Reducing this error is therefore a question of what the negative class covers, not of the threshold it is checked against. Widening the `none` class with more varied non-signing motion, particularly the kind of motion these thirteen clips resemble, is the direct way to give the classifier a better boundary to place, and it is the single most useful addition to this evaluation.

5.14.4 Duration Is Only Encoded Below the Sequence Length

The model has no explicit time axis and reads frame count as duration, which is why the server discards frames down to the capture rate of the training data. That reasoning holds for clips shorter than 137 frames, which is 797 of the 840. For the 41 longer clips, uniform subsampling compresses the performance to look exactly 137 frames long, so their duration information is removed. Anything longer than about 8.7 seconds of signing is therefore scale-normalised and not measured. This is an acceptable engineering choice at this vocabulary size, and it would need revisiting for longer phrases.

A related effect follows from the masking layer. A frame in which MediaPipe detected neither the pose nor either hand is all zeros, and is therefore indistinguishable from padding and skipped. A clip containing many undetected frames is consequently shorter, from the model's point of view, than its frame count suggests.

5.14.5 The Network Dependency

Recognition runs on the server, so the delivered system does not work without a network connection. For a tool intended for emergencies this is a real weakness and not a neutral trade, and it is the first item in the future work of Chapter 6. The reason it was accepted is given in Section 3.9.1: the alternative was to reimplement the feature pipeline against a different landmark source, where a numerical mismatch produces confident wrong answers rather than an error, and the project had no measurement establishing that the two sources agree.

CHAPTER 6
CONCLUSION

6.1 Conclusion

This project built a working recogniser for six Nepali Sign Language emergency phrases and delivered it as an application that displays and speaks the recognised phrase.

The signed form of every phrase was established in person with Deaf experts at the Rehabilitation Centre for the Rural Deaf and the National Federation of the Deaf Nepal before any data was recorded, so the system recognises real NSL and not an approximation of it. Eight signers then contributed 840 clips, balanced at exactly 120 per class across the six phrases and a negative class for non-signing motion. No public dataset of NSL emergency phrases existed at the start of this work, and this one is the project's most durable output.

Representing each frame as 225 pose and hand landmarks instead of pixels is what made a dataset of this size sufficient. Normalising against the shoulder midpoint and each wrist removes the signer's position and distance from the camera, and standardising every clip to 137 frames, the 95th percentile of the measured frame-count distribution, gives the classifier a fixed input without discarding the shape of the motion. A bidirectional LSTM of 192,007 parameters classifies the result and trains in minutes on a laptop processor.

Under leave-one-signer-out evaluation the system reaches 96.52% mean accuracy across eight folds and 96.67% pooled, with per-class F1 between 0.963 and 0.996 for the six emergency phrases. The same architecture reaches 99.21% under a random split, and reporting both is deliberate: the difference of 2.5 percentage points is a measurement of how much a random split overstates performance when clips from one person can appear on both sides of it. Published NSL recognition results are generally reported without this separation, so the lower figure here is the more comparable one to any future work that adopts the same protocol.

Because a six-way classifier would name a phrase for any input at all, the system trains a seventh, negative class on non-signing motion and requires the softmax confidence to clear a threshold before announcing a phrase. For a tool whose output is a spoken emergency announcement, the ability to stay silent is a feature of the same standing as accuracy.

The trained model was exported to TensorFlow Lite, verified against the original to a largest probability difference of 1.2×10^{-7}, and deployed behind a stream-

ing inference server with an Android client. Recognition works end to end on a physical device.

6.2 Limitations

1. **The system requires a network connection.** Recognition runs on the server, not the phone. The proposal stated that no server would be needed at use time, and the delivered system does not meet that. For an emergency tool this is the most consequential limitation in this list, and Section 3.9.1 gives the reason it was accepted.

2. **Eight signers, four of them the authors.** Every accuracy reported rests on eight people. The standard deviation across folds is 2.41 percentage points and the range is seven, so the estimate of performance on a new signer carries real uncertainty. Four of the eight are not fluent NSL users; they reproduced the consultants' reference performance. The reported accuracy is therefore an accuracy on a mixture of fluent and learned signing.

3. **Six phrases.** The vocabulary is fixed and small. The system cannot express anything outside it, and a user with a need not on the list has no way to convey it.

4. **The negative class is the weakest class.** Recall on `none` is 0.892, and thirteen non-signs were classified as emergency phrases, each a case where the system would announce an emergency that nobody signed. All thirteen cleared the confidence threshold, so the fix is a broader negative class and not a stricter threshold.

5. **Speech output is in English.** The client uses the device's text-to-speech voice, and Nepali is absent from most Android devices. The proposal specified pre-recorded Nepali audio, which was not implemented.

6. **Duration is only encoded below 137 frames.** Clips longer than the sequence length are uniformly subsampled, so signing longer than about 8.7 seconds is scale-normalised rather than measured.

7. **Facial expression is ignored.** The consultants confirmed no phrase in this set depends on it, so the exclusion is safe for these six phrases and would not hold for a larger vocabulary.

8. **One sign at a time.** The system recognises a single complete phrase per interaction and does not segment continuous signing or handle more than one person in frame.

6.3 Future Work

1. **Move inference onto the device.** This would remove the network dependency, the most serious limitation above. The obstacle is that the dataset was recorded with MediaPipe Holistic, which exists only as a Python pipeline, and an on-device implementation would use a different landmark source producing slightly different numbers. The comparison needed to settle whether a model trained on one source survives the other is implemented in the codebase but has not been run; it requires recording some 35 clips through both paths at once. That measurement is the correct next step, since it either unblocks on-device inference or establishes that re-recording the dataset is the price of it.

2. **Broaden the negative class.** Recording more varied non-signing motion, targeted at the kind of motion the thirteen misclassified clips resemble, is the direct route to reducing false announcements.

3. **Recruit more signers, prioritising fluent ones.** The clearest route to a more trustworthy accuracy figure is more people, and specifically more fluent NSL users. Doubling the signer count would narrow the confidence in the leave-one-signer-out estimate more than any change to the model. Section 5.8 supports that ordering with a measurement rather than an argument: seven architectures spanning a sevenfold range of parameter counts differ from one another by less than the spread between signers, so architecture search has little left to give while signer coverage has a great deal.

4. **Record Nepali audio.** The phrase set is fixed at seven, so seven recordings by a fluent speaker would replace the English text-to-speech entirely.

5. **Extend the vocabulary.** The pipeline is not specific to these six phrases. Adding a phrase requires the consultation step, recordings, and a retrain, with no change to the preprocessing or the architecture.

6. **Handle continuous signing.** Recognising a phrase within a stream, with no taps to mark its boundaries, would remove the need for the user to mark the boundaries and is the main step from a proof of concept toward something usable without instruction.

7. **Reduce the confusion between `call_police` and `cant_breathe`.** Four of the six phrase-to-phrase errors are this one pair, which makes it a specific and tractable target.

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APPENDIX A

Field Consultation with NSL Experts

The signed form of each of the six emergency phrases was established in person with Deaf experts before any data collection began, as described in Section 3.2. Consultations were held at the Rehabilitation Centre for the Rural Deaf in Bhaktapur and at the National Federation of the Deaf Nepal in Ranibari Marga, Samakhushi, Kathmandu. The images below are reproduced with the consent of those photographed.

Figure A.1: The project team with the NSL consultant following a gloss consultation session

Figure A.2: Reviewing a recorded demonstration with the consultant to confirm the reference form of a phrase

Figure A.3: An NSL expert demonstrating the signed form of a phrase for reference recording

APPENDIX B

Data Collection Tool

Clips were recorded with a purpose-built tool that runs MediaPipe Holistic on a live webcam feed and saves the 225-value landmark vector for each frame. The operator selects a signer identity and a phrase, then starts and stops each recording explicitly. Landmark overlays are drawn live, so a failed detection is visible during recording instead of afterwards.

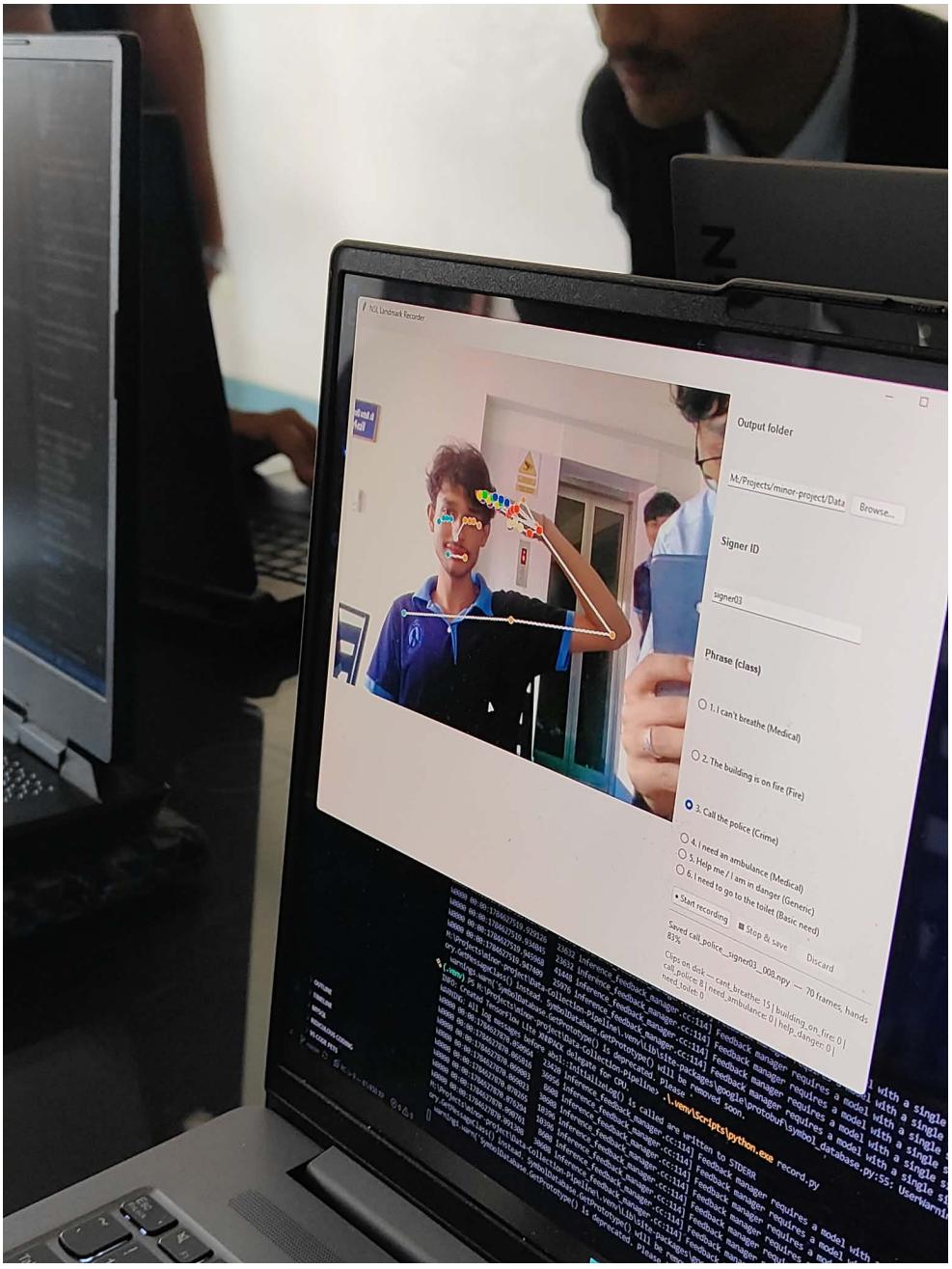

Figure B.1: A signer recording the *Call the police* phrase, showing live pose and hand landmark overlays alongside the signer identity and phrase selection

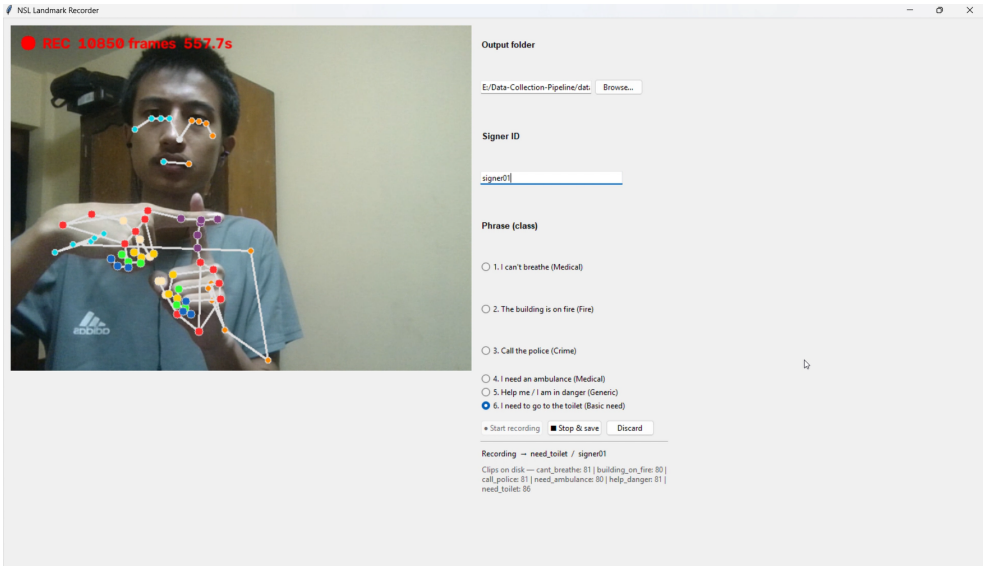

Figure B.2: The recorder during an active recording, showing hand landmarks in detail and the running per-class clip counts on disk

APPENDIX C

Dataset Organisation

The eight signers recorded independently on separate machines. Contributions were merged through a shared Git repository instead of by hand, which gave a reviewable history of what each signer added and caught duplicate and misnamed clips before they entered the combined dataset.

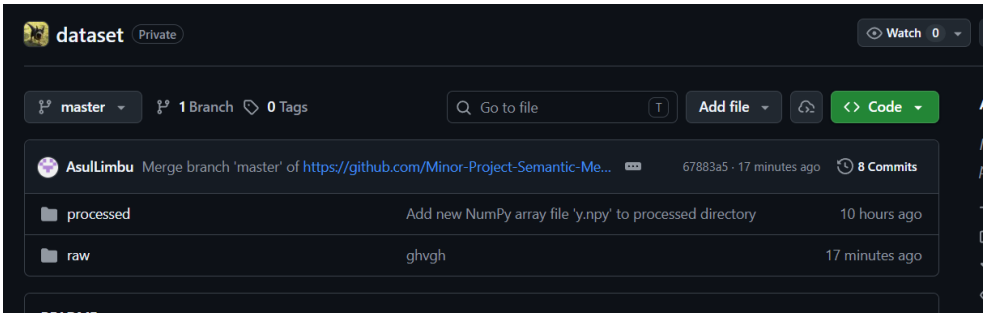

Figure C.1: The shared dataset repository, with raw and processed directories tracked separately and a commit history showing contributions merged from several signers

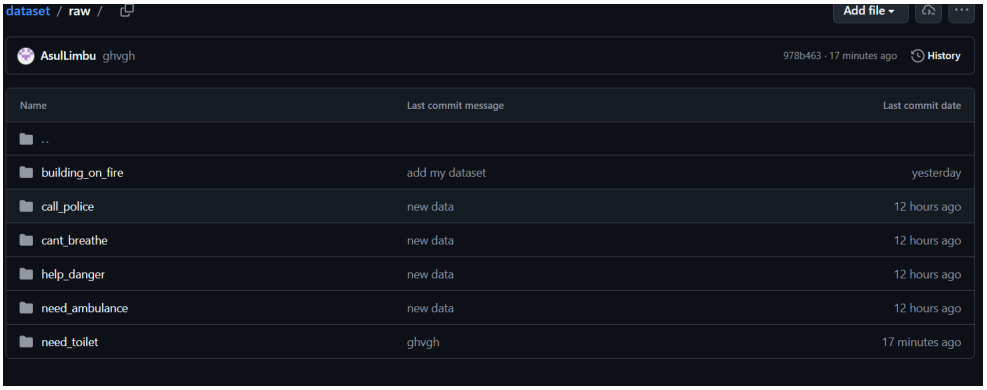

Figure C.2: The raw directory after merging, with one subfolder per phrase class holding the contributions of all signers

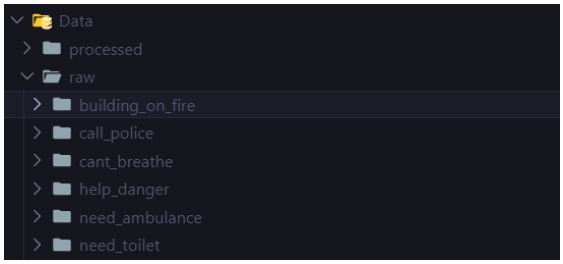

Figure C.3: Local directory layout: raw clips organised by phrase class, alongside the processed output directory

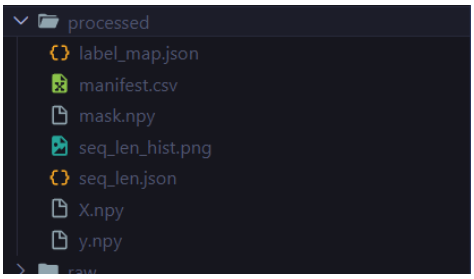

Figure C.4: Contents of the processed directory: the compiled tensors alongside the manifest and the sequence-length metadata

APPENDIX D

Sample Landmark Data

This appendix traces one clip through the preprocessing pipeline, to make the numeric effect of each step concrete. The clip is `call_police__signer01__001.npy`, stored as a $(42, 225)$ array of 32-bit floats: 42 recorded frames, each holding one 225-value landmark vector.

D.1 Feature Vector Layout

Table D.1: Layout of the 225-value feature vector

Block	**Indices**	**Values**	**Content**
Pose	0 to 98	99	33 landmarks $\times$ (x, y, z)
Left hand	99 to 161	63	21 landmarks $\times$ (x, y, z)
Right hand	162 to 224	63	21 landmarks $\times$ (x, y, z)
Total		**225**	

D.2 Raw Values

The first five pose landmarks of the clip's first frame, in the normalised image coordinates MediaPipe reports:

Table D.2: Raw pose landmarks 0 to 4, first frame

Landmark	x	y	z
0	0.3683	0.3315	-1.0291
1	0.3924	0.2687	-0.9459
2	0.4069	0.2707	-0.9461
3	0.4187	0.2732	-0.9460
4	0.3420	0.2707	-0.9640

Both hand blocks of this frame are entirely zero. The signer's hands had not yet entered the frame when recording began, MediaPipe reported no hand landmarks, and the recorder filled all 126 values with zeros instead of leaving them undefined.

D.3 After Normalisation

The same five landmarks after dual-anchor normalisation, now expressed relative to the shoulder midpoint and scaled by shoulder width:

Table D.3: Normalised pose landmarks 0 to 4, first frame

Landmark	x	y	z
0	−0.0054	−0.7148	−1.6152
1	0.0463	−0.8491	−1.4370
2	0.0772	−0.8449	−1.4374
3	0.1025	−0.8396	−1.4374
4	−0.0617	−0.8449	−1.4758

The x coordinates now sit around zero instead of around 0.37, because the origin has moved to the midpoint of the shoulders. The zero hand blocks remain exactly zero, since both normalisation formulas map a zero input to a zero output.

D.4 After Length Standardisation

The clip has 42 frames, fewer than the sequence length of 137, so it is zero-padded at the end. The result is a $(137, 225)$ array in which the first 42 frames are real and the remaining 95 are zero, with a boolean mask recording which is which. The final frame of the standardised clip is entirely zero, confirming that padding stays distinguishable from signal and that the masking layer will skip it.

APPENDIX E

Client Interface

The Android client presents one of two outcomes for every recognition attempt: an accepted phrase, which is displayed and spoken, or a low-confidence result, which is reported without being spoken. A row of indicators below the camera preview reports what the server observed for each frame, so a failure during use can be attributed instead of guessed at.

[**Insert client screenshots here: the accepted and low-confidence result states, the telemetry indicator row, and the server configuration screen.**]

APPENDIX F

Server Interface

The inference server exposes the endpoints in Table F.1. The streaming endpoint is the path the client uses; the two prediction endpoints exist so that the server can be checked against the offline pipeline without needing a phone.

Table F.1: Server endpoints

Method	**Path**	**Key**	**Purpose**
GET	`/health`	no	Backend, class list, thresholds, target frame rate
GET	`/classes`	yes	Class keys and display labels
POST	`/predict/landmarks`	yes	A clip supplied as JSON, for verification
POST	`/predict/npy`	yes	The same clip as raw array bytes
WS	`/ws/stream`	yes	Live frames to a decision, the path the client uses

The health endpoint is deliberately unauthenticated so that container health checks and the client's reachability test keep working when a key is configured. Every other endpoint requires the key, compared with a constant-time function so that a wrong key cannot be narrowed down by timing the response.

Table F.2: Messages on the streaming endpoint

Direction	**Message**	**Meaning**
Server	`ready`	Session established; carries the target frame rate and class list
Client	binary frame	One JPEG frame
Server	`ack`	Whether the frame was kept, the running frame count, and whether pose and hands were detected
Client	`done`	Signing has finished; produce a decision
Server	`result`	Status, phrase, confidence, and the three most probable classes
Client	`reset`	Discard the current clip and start again
Client	`ping`	Keepalive, sent while idle between signs